

Respiratory Physiology

Outline

1) Introduction

Functional structures of the respiratory system

2) Pulmonary ventilation

Mechanics of breathing

3) Pulmonary gas exchange and Tissue gas exchange

4) Gas transport in the Blood

5) Respiratory Regulation

6) Respiratory insufficiency

INTRODUCTION

1. General consideration

1.1. The essentials for survival:

- a. Integrity of physiological systems**
- b. Supply of extrinsic materials including:**
 - ❖ food,
 - ❖ water
 - ❖ and air

Introduction cont'd.....

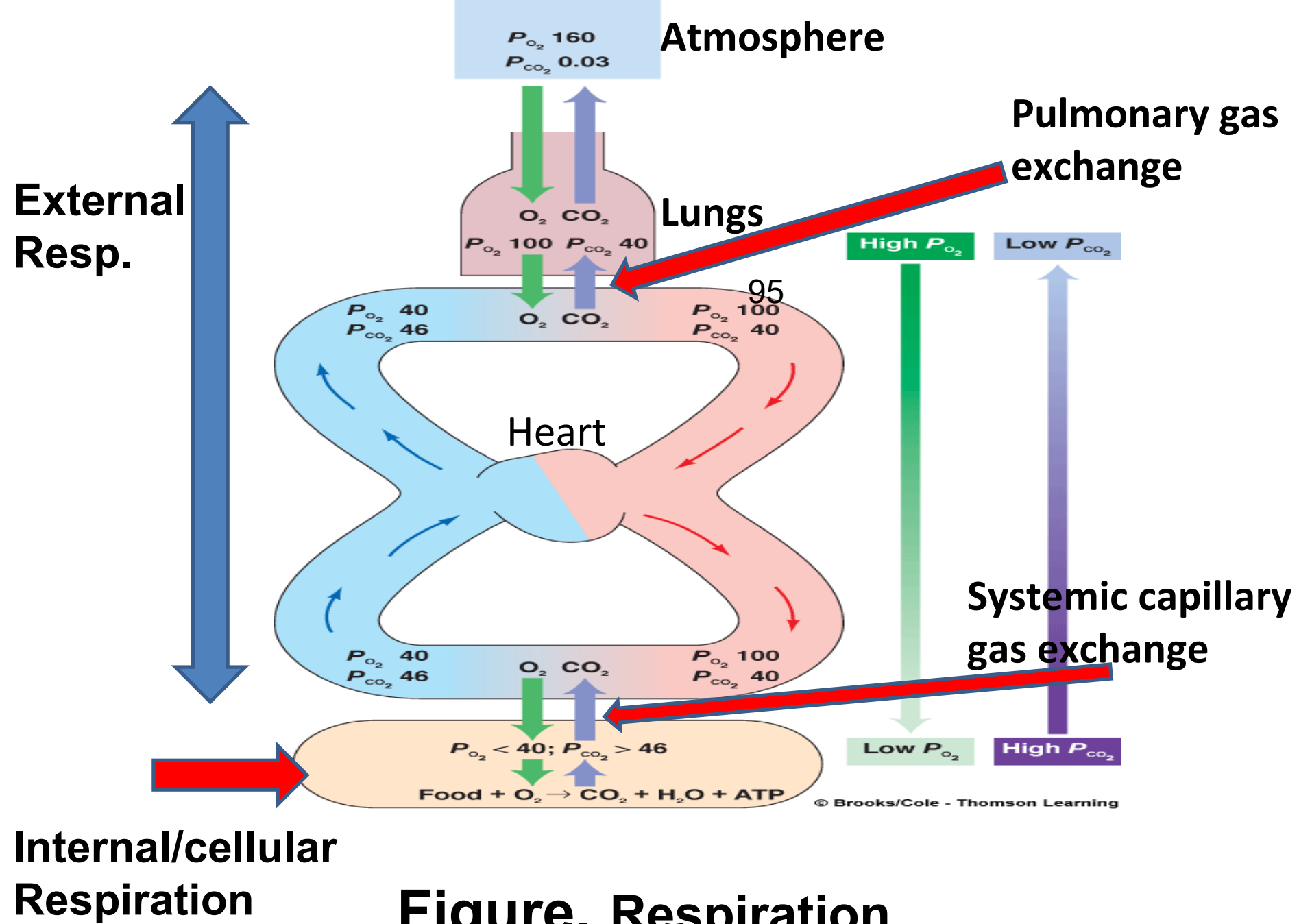
1.2. Exchange of gases in animals in general

- **Unicellular organisms:** Simple diffusion
- **Large animals:** the Prime function of the lung is to allow exchange of oxygen for carbon dioxide .
- The lung also metabolizes some compounds, filters unwanted materials from the circulation, and acts as a reservoir for blood.

Introduction cont'd.....

Respiration:

- Is the process of taking up oxygen and removing carbon dioxide from cells in the body
- There are two phases of respiration:
 - 1) External respiration** is the exchange of O_2 and CO_2 between the blood and the alveoli
 - 2) Internal respiration/cellular respiration** is a series of complex metabolic reactions that break down molecules of food, releasing water, carbon dioxide and **energy**



Stages of Respiration

- 1. Pulmonary ventilation: gas exchange between the atmosphere and lungs**
- 2. External respiration -gas exchange between the lungs and blood (O_2 loading and CO_2 unloading).**
- 3. Transport of respiratory gases –via movement of blood O_2 from the lungs is transported to the cell and tissues.**
- 4. Internal respiration –gas exchange between the capillaries and the tissues (O_2 unloading and CO_2 loading).**

The functions of the respiratory system:

- Gas conduction
- Gas exchange

This is called **respiratory function** of the respiratory system

There are also **non-respiratory functions** including:

- Provision of a route for water and heat elimination
- Olfaction
- Maintenance of acid base balance
- Vocalization (phonation)
- Defense (alveolar macrophages, ciliary beating, cough reflex, mucus secretion, etc)
- Metabolism (synthesis/modification/activation/ inactivation of molecules; example angiotensin II)
- Enhances venous return
- Blood reservoir

Functional structures of respiratory system

The **respiratory system** consists of the nose (mouth), pharynx (throat), larynx (voice box), trachea (windpipe), bronchi, and the lungs

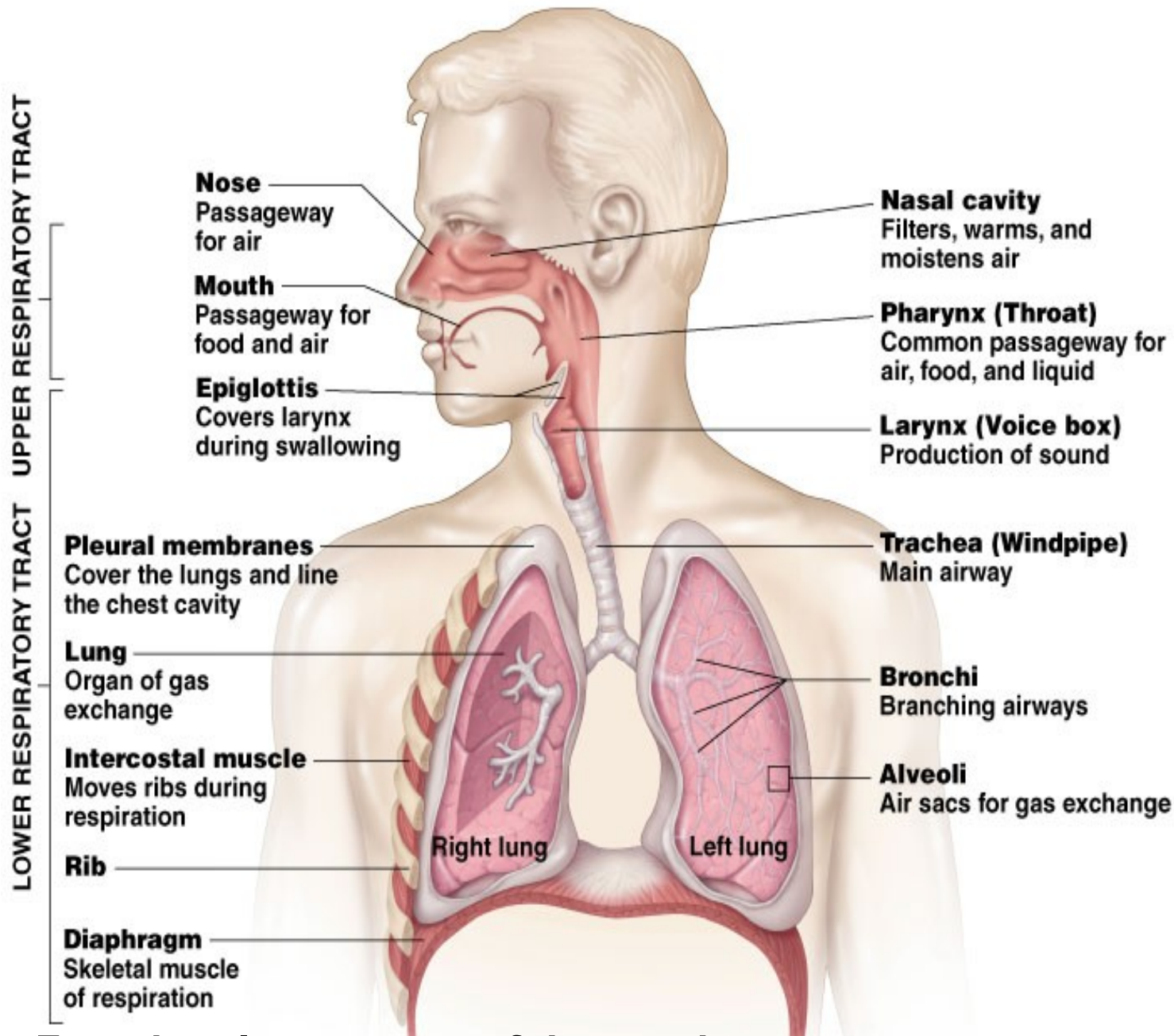


Figure 2. Functional structures of the respiratory system

Structurally (anatomically)-the respiratory system consists of two parts

(1) **The upper respiratory system**
includes the nose and pharynx

(2) **The lower respiratory system** includes the
larynx, trachea, bronchi, and lungs

Functionally (physiologically)- the respiratory system also consists of two parts:

(1) Conducting zone and (2) Respiratory zone

(1) **The conducting zone:**

- Consists the nose (mouth), pharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles
- Their function is to filter, warm, and moisten air and **conduct air into and out of the lungs**

(2) The **respiratory zone**

- consists respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli
- sites of gas exchange between air and blood

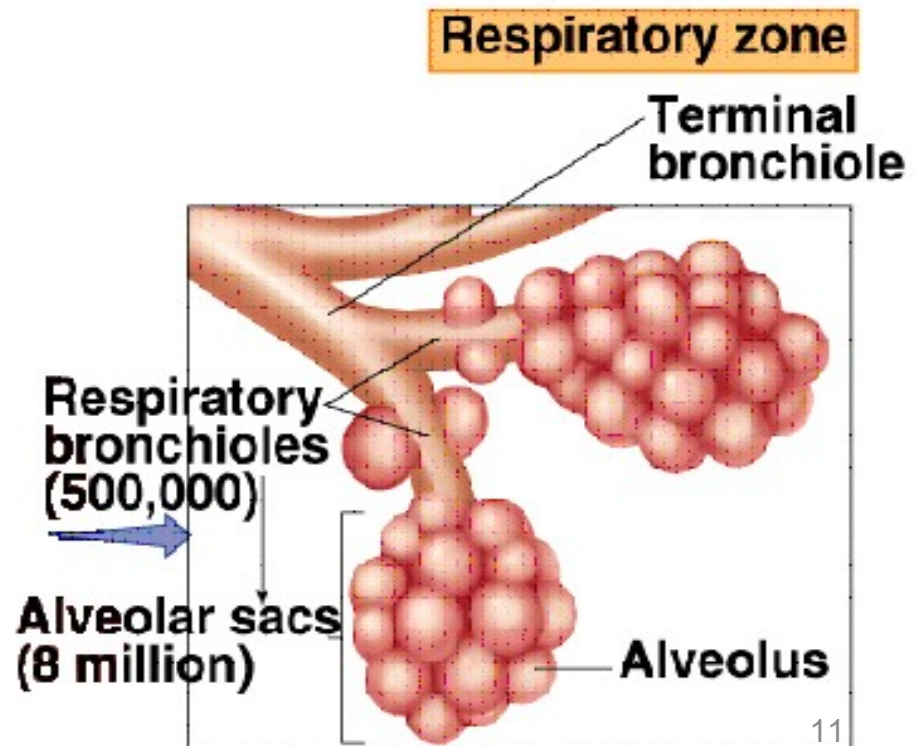


Fig. 3. Respiratory zone

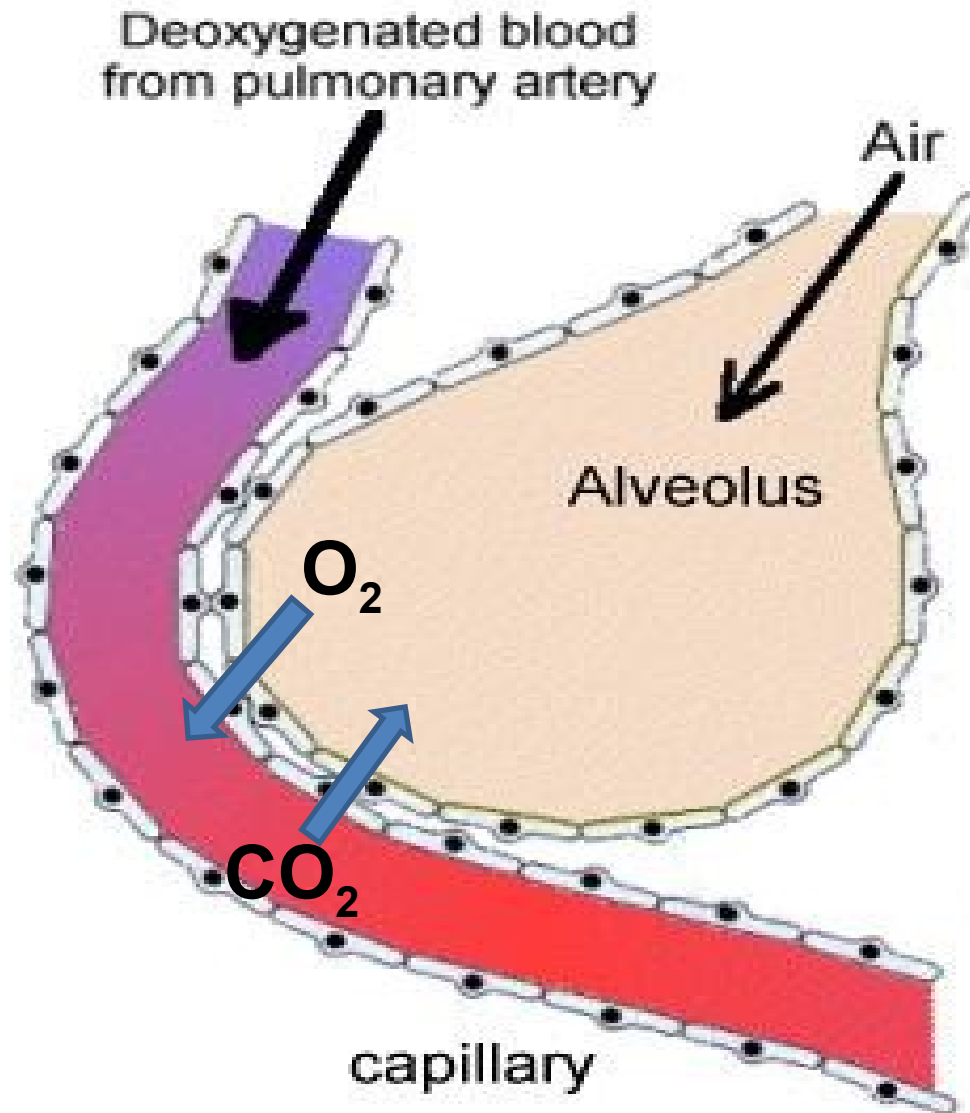


Fig. 4 Gas exchange in pulmonary capillary

The Upper Airway Conducting Zone

1. The Nasal Cavity consists of:

The is the most protruded structure in the face.

Nasal septum: partition between right and left halves of the nasal cavities

Nostrils: two openings in the nasal cavities through which air enters; also called anterior nares

The nose also has posterior nares opening into the pharynx.

Vestibules (the first ~1.5 cm of the conductive portion following the nostrils) lined with a keratinized stratified squamous epithelium.

Nasal conchae (turbinates): three bony projections on the lateral walls of the nasal cavities; increase the surface area of the nasal cavity and create turbulence in the stream of passing air, both of which facilitate the conditioning (warming, moistening and filtration) of the air.

- Contains
- Paranasal sinuses: drainage openings
- Olfactory organs: receptor cells for the sense of smell
- Sebaceous glands contribute to humidification of inspired air.

2. The Pharynx breaks into:

a. **Oesophagus** used for food and fluid passage.

b. **Larynx** extends from C3 to C6 for air conduction and consists of:

- Thyroid cartilage
- Epiglottis
- Vocal cords
- Glottis

The larynx connects the pharynx to trachea and serves as first portion of air passage.

Its vocal folds provide: control of airflow and production of voice.

It Provides a patent airway

Acts as a switching mechanism to route air and food into the proper channels

It contracts rapidly, to prevent food and liquids from entering the airway.

The Lower Airway Conducting zone

Trachea

Bronchi

Bronchioles

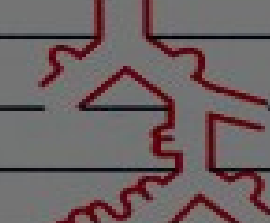
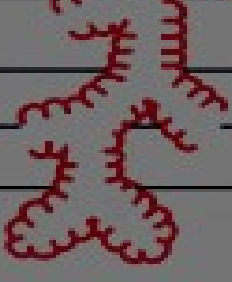
TRACHEA

A fairly short tube (10-12 cm) long with a diameter of about 2cm.

Lined by respiratory epithelium.

Extends from the larynx into the mediastinum

Musculo-cartilaginous structure that branches into two bronchi each of which breaks into 23 generations of progressively smaller bronchi, terminal bronchioles and respiratory bronchioles.

Conducting zone	Trachea		0
	Bronchi		1
			2
			3
			4
	Bronchioles		5
	Terminal bronchioles		16
Transitional and respiratory zones	Respiratory bronchioles		17
			18
			19
	Alveolar ducts		20
			21
			22
	Alveolar sacs		23

Idealization of the human airways according to Weibel. Note that the first 16 generations (Z) make up the conducting airways, and the last 7, the respiratory zone (or the transitional and respiratory zone).

BRONCHI (Bronchial Tree)

Conductive structures of a size down to ~1 mm are termed bronchi.

Bronchi are extension of trachea making the 1st to 11th generations while the terminal bronchioles make up the 12th-16th generations.

At the transition from bronchi to bronchioles the epithelium changes to a ciliated columnar epithelium

Respiratory bronchioles with alveoli in their walls make up the 17th-20th generations;

Alveolar ducts making up the 21st-23rd generations terminate in alveolar sacs.

The Cilia

- Hair-like projections lining the respiratory passages to the end of the bronchioles
- When one breathes, one essentially inhales a lot of junk. The ciliary system is required to remove the junk or particulate matter sticking to mucus.
- Particulate matter including micro-organisms are moved by cilia to pharynx and swallowed thus keeping the lungs clean.
- The mucus in cilia forms an “escalator” called mucus escalator. This facilitates movement of particulate matter and thus defends the air passages against bacterial infection.
- Ciliary activity reduced by noxious agents leads to lung infection.
- For example, a single cigarette smoked → non-motility of cilia for several hours
- Non-motility of cilia + secretion of mucus → accumulation of secretion → partial or complete airway obstruction.

Alveolar duct is a branch of a respiratory bronchiole around which alveoli and alveolar sacs are arranged

Alveolus is a small hollow or cavity, a cup shaped out pouching lined by simple squamous epithelium and supported by a thin elastic basement membrane

Alveolus is the site of gas exchange in the lungs

Alveolar sac is a cluster of alveoli that share a common opening

The lungs

They form the surface (respiratory membrane) over which **oxygen is absorbed in to blood** and **carbon dioxide is excreted from blood**

Each lung is enclosed and protected by a double-layered serous membrane called the **pleural membrane**

The **parietal pleura**, lines the wall of the thoracic cavity; the deep layer, the **visceral pleura**, covers the lungs themselves

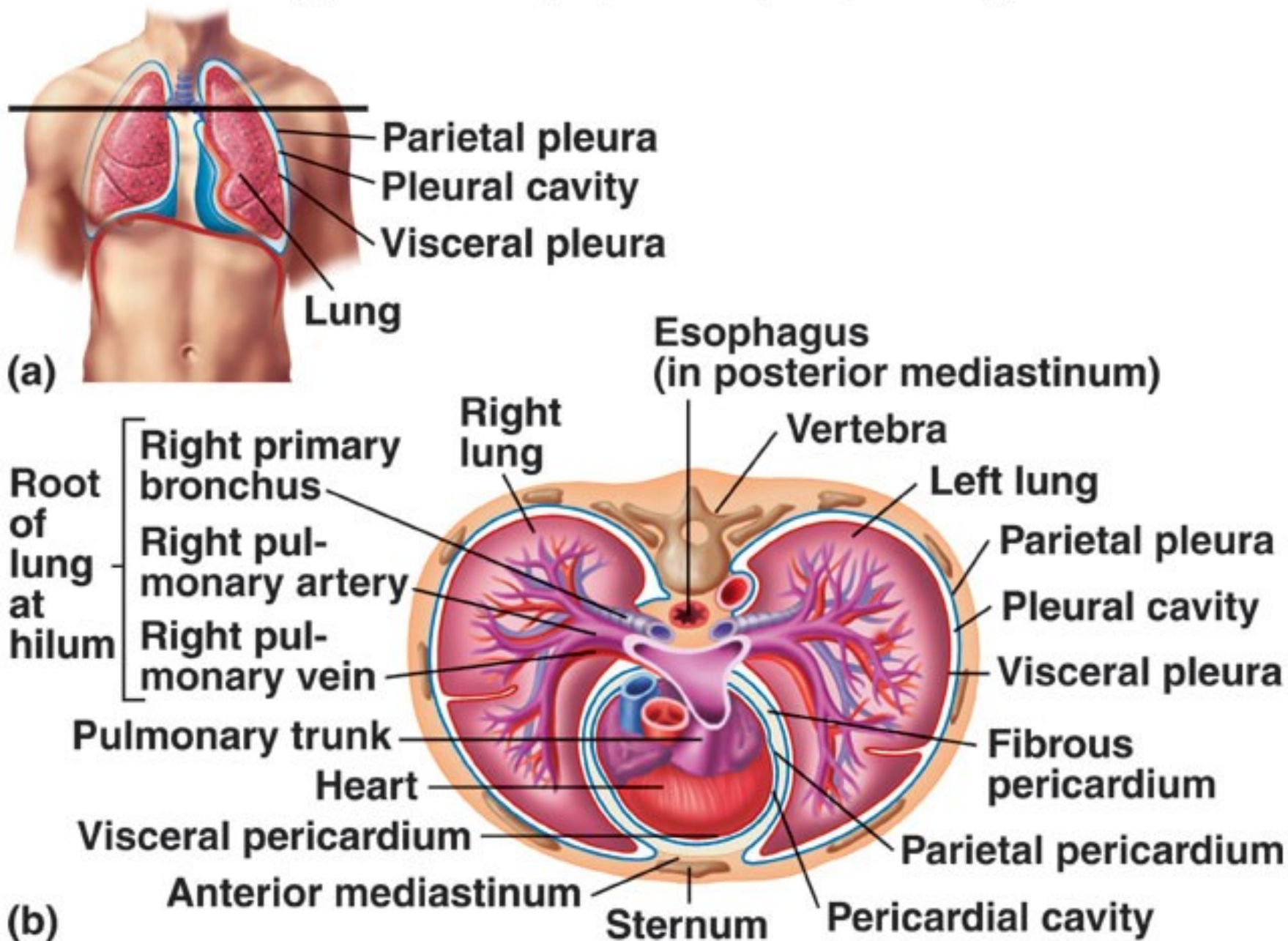


Fig. 7 Pleural membrane

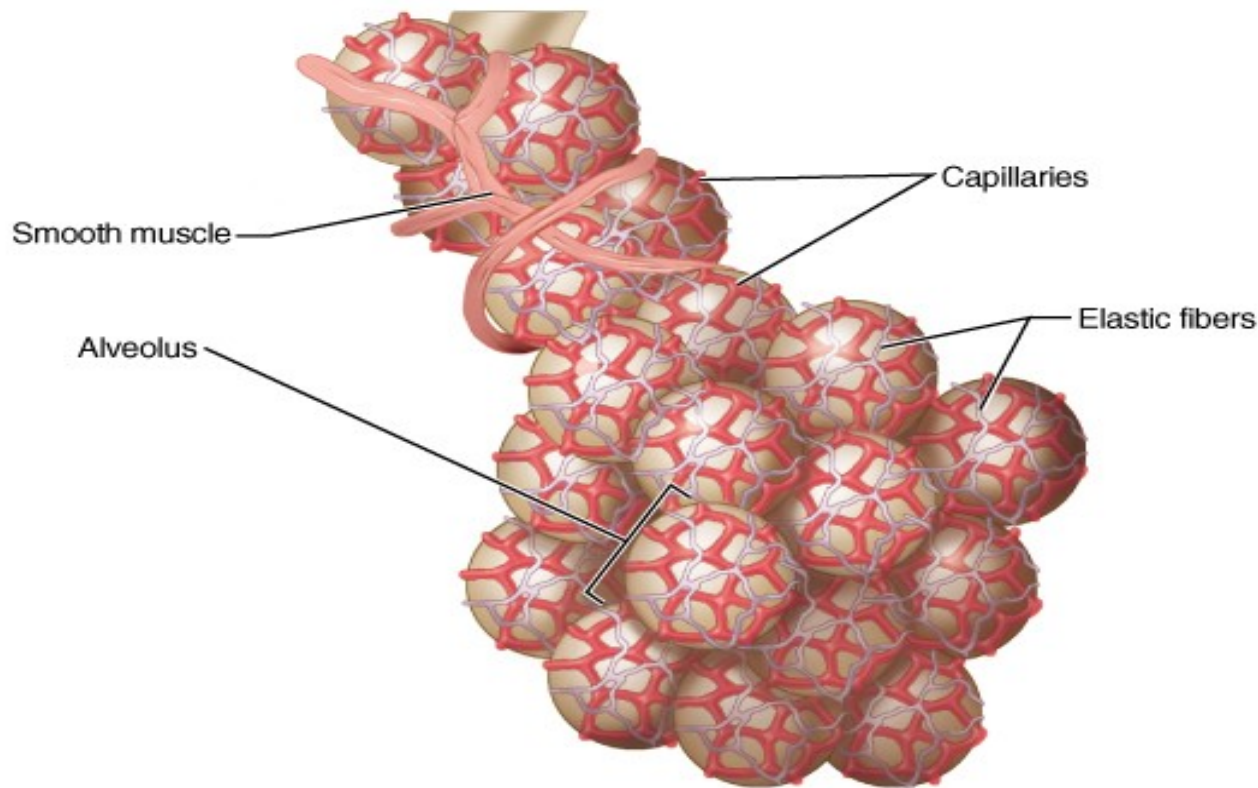
Between the visceral and parietal pleurae is a small space, the **pleural cavity**, which contains a small amount (which is about 10 ml.) of **lubricating** fluid secreted by the membranes (**pleural fluid**)

Pleural fluid also causes the two membranes to **adhere** to one another

Lungs contain numerous alveoli

An alveolus is a small hollow or cavity, an air sac in the lungs

There are about **300 million alveoli** in the adult lungs and each alveolus is almost completely enveloped by pulmonary capillaries



(b)

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Fig. 8 Alveoli

Each alveolus is 1 cell layer thick

Alveoli have walls made up of pneumocytes of three types:

Type I pneumocytes- major lining cells, where gas exchange takes place;

Type II pneumocytes- that are less in number and constitute thicker granulocytes responsible for the production of **surfactants that lower surface tension & prevent lung collapse**

Type III pneumocytes- large phagocytic macrophage cells found in the alveolar cavities.

These cells keep **alveolar surfaces sterile by removing debris and microbes**

Are supplied with blood vessels, lymphatics.

Are connected by open pores

Allow air pressure throughout the lung to be equalized.

Form part of the respiratory (alveolo - capillary) membrane.



Alveolar structure

The alveoli are composed of type I cells for gas exchange and type II cells that synthesize surfactant. Alveolar macrophages ingest foreign material that reaches the alveoli.

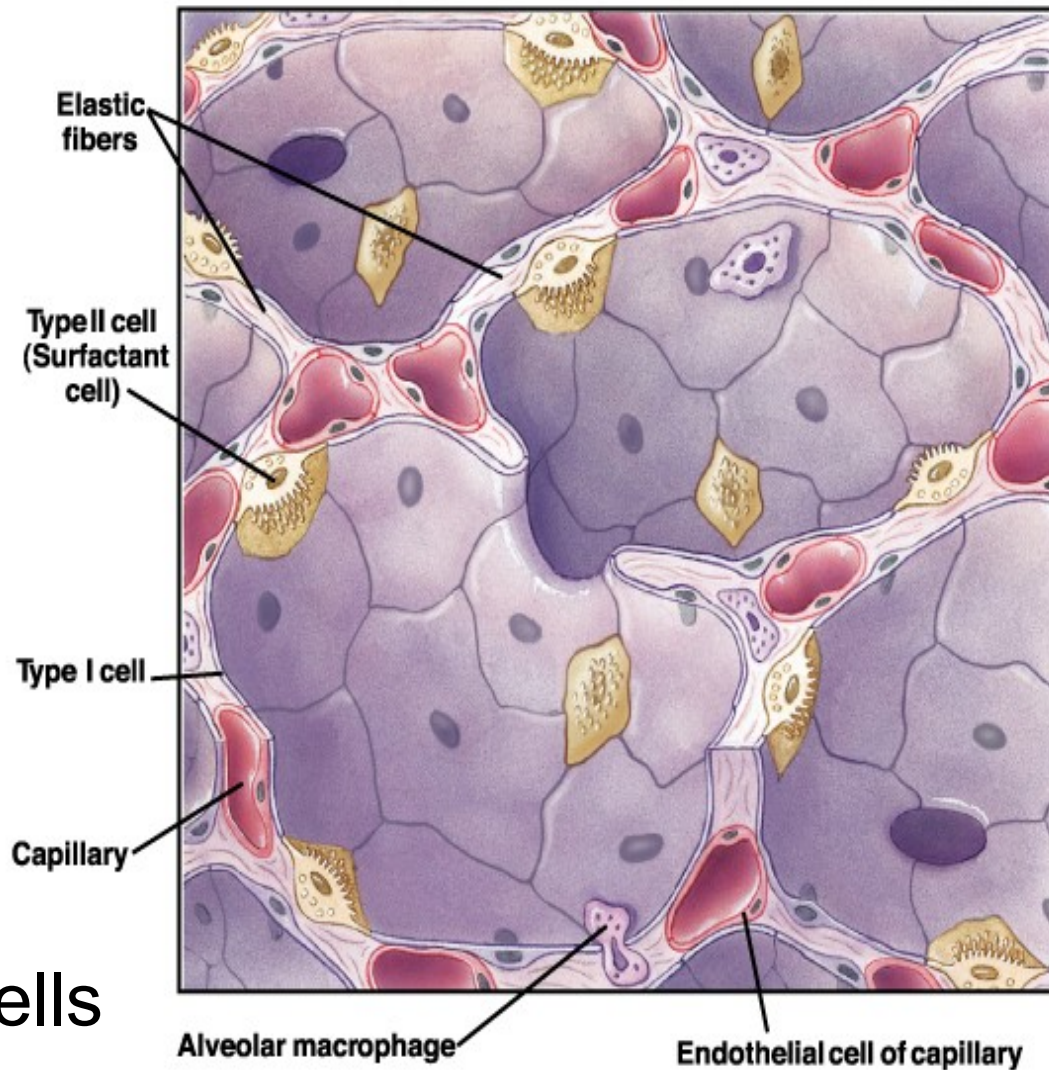


Fig. 9. Alveolar cells

Air filtering function of respiratory passageways

Three distinct functions are performed on air being inhaled:

- (1) the air is *warmed*
- (2) the air is *humidified*
- (3) the air is *filtered*

These functions together are called the *air conditioning function* of the respiratory passage ways

Resistance to Airflow in the Bronchial Tree

Under *normal respiratory conditions*:

- Air flows through the respiratory passageways **so easily** that less than 1 centimeter of water pressure gradient from the alveoli to the atmosphere is sufficient to cause enough airflow for quiet breathing
- Greatest amount of resistance to airflow occurs in some of the **larger bronchioles and bronchi near the trachea** (but not in the minute air passages of the terminal bronchioles)

Under disease conditions the smaller bronchioles often play a far greater role in determining airflow resistance because of their small size they are easily occluded by muscle contraction in their walls due to:

- (1) Edema occurring in the walls, or
- (2) Mucus collecting in the lumens of the bronchioles

Nervous and Local Control of the Bronchial Musculature

Sympathetic nerve fibers:

- Control of the bronchioles is relatively weak because few of these fibers penetrate to the central portions of the lung
- But bronchial tree is very much exposed to *norepinephrine* and *epinephrine* released into the blood by sympathetic stimulation and adrenal gland medullae; both cause dilation of bronchial trees (β_2 adrenergic receptors)

Parasympathetic nerve fibers (vagus nerves):

- few parasympathetic nerve fibers penetrate the lung parenchyma
- cause mild to moderate constriction of the bronchioles (acetylcholine)

Local secretory factors:

- several substances formed in the lungs are often quite active in causing bronchiolar constriction
- most important of these are *histamine* and *slow reactive substance of anaphylaxis*
- these are released in the lung tissues by *mast cells* during allergic reactions
- they play key roles in causing the *airway obstruction that occurs in allergic asthma*

The structure of the chest wall

The lungs and airways alone cannot bring air in and out of the lungs or exchange oxygen and carbon dioxide from the blood

Muscles and an airtight chest wall to create a negative pressure within the chest are important to move air into/out of the lungs

Blood flow to the lungs, and matching blood perfusion and airflow are essential for gas exchange

Thus the respiratory system works in coordination with other systems in the body to provide O_2 to tissues and to remove CO_2 from the body

The lung is an elastic structure that collapses like a balloon and expels its air through the trachea whenever there is no force to keep it inflated

Inflation of the lungs is achieved by changing the dimensions of the chest wall by means of the respiratory muscles

Pulmonary ventilation

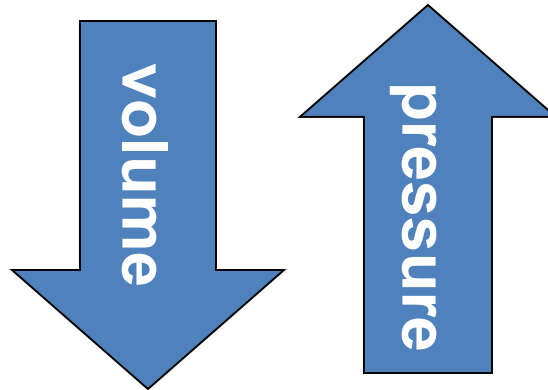
Pulmonary ventilation is the inflow and outflow of air between the atmosphere and the lung

Mechanics of Pulmonary Ventilation

The action of breathing in and out is due to changes of pressure within the thorax and the lungs, in comparison with the outside (atmosphere) pressure

In accordance with *Boyle's Law*, when the volume of a given quantity of gas increases its pressure decreases and vice versa at constant temperature:

$V \propto \frac{1}{p}$; Volume is inversely proportional to pressure



Under resting respiration, contraction of respiratory muscles causes expansion (increase) in thoracic cavity volume while relaxation of the respiratory muscles causes decrease in thoracic cavity volume

Change in thoracic cavity volume is accompanied by change in pressure in the thoracic cavity (intrapleural pressure)

Change in intrapleural pressure in turn causes change in alveolar pressure

Air flows from area of higher pressure to area of lower pressure

Pulmonary ventilation is brought by changing intrapulmonary pressure (alveolar pressure) to subatmospheric pressure and above atmospheric pressure during inspiration and expiration respectively

The lungs can be expanded and contracted /lung pressure can be changed/ in two ways by changing chest wall volume:

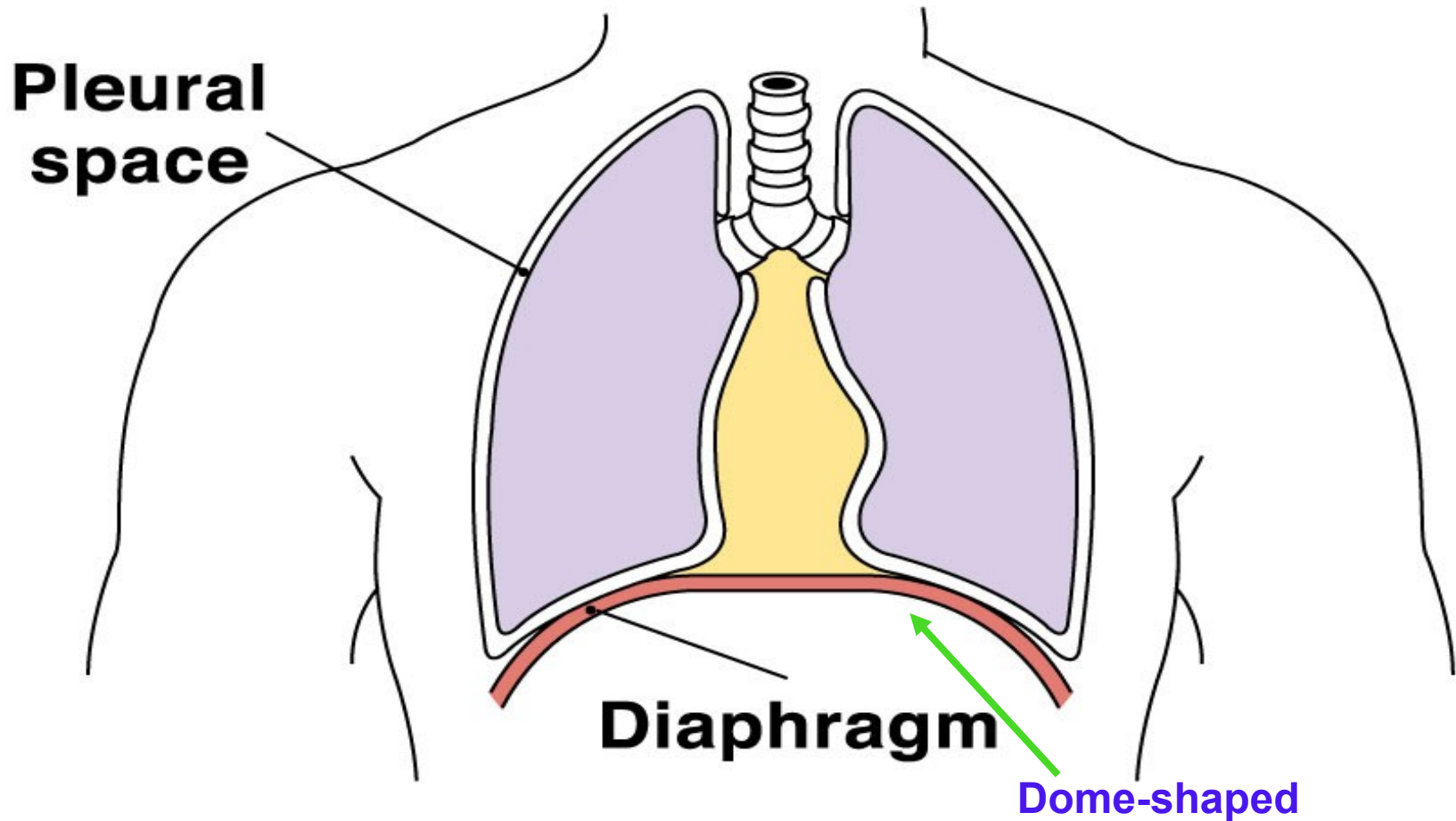
- (1) by downward and upward movement of the diaphragm to lengthen or shorten the chest cavity, and
- (2) by elevation and depression of the ribs to increase and decrease the anteroposterior diameter of the chest cavity

Normal quiet breathing is accomplished almost entirely by **movement of the diaphragm down**

Diaphragm is a thin dome-shaped sheet of muscle that is inserted into the lower ribs

It is supplied by the phrenic nerves

Diaphragm is the main muscle of inspiration

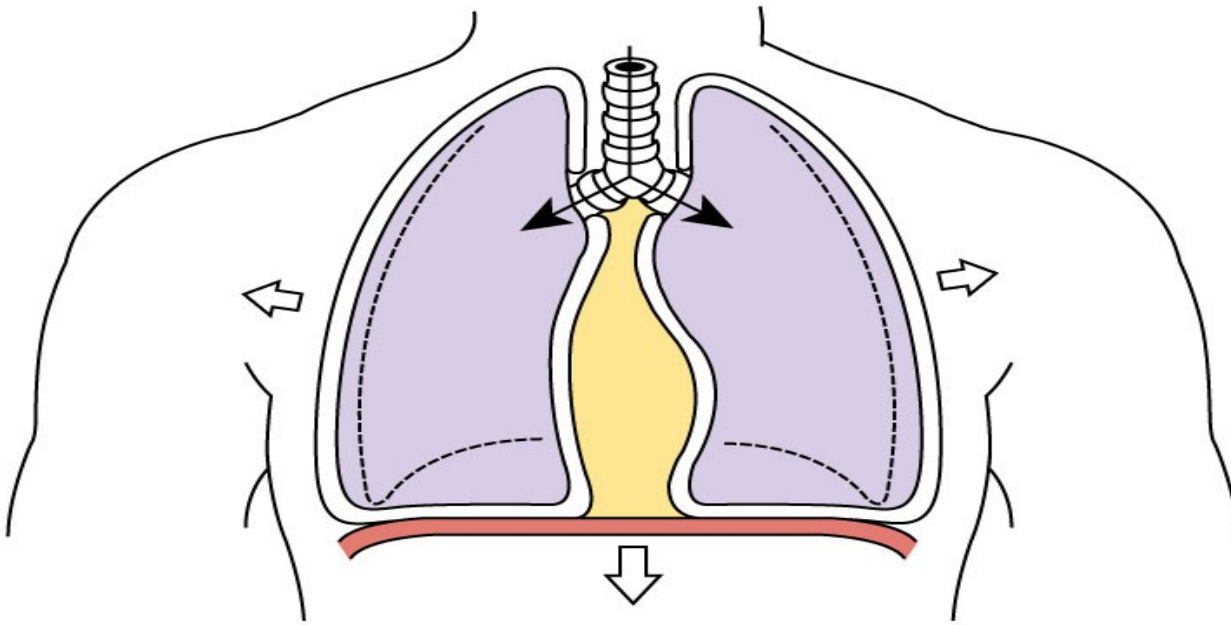


At rest, diaphragm is relaxed

Fig. 10 Diaphragm

Inspiration

- the vertical dimension of the chest cavity is increased



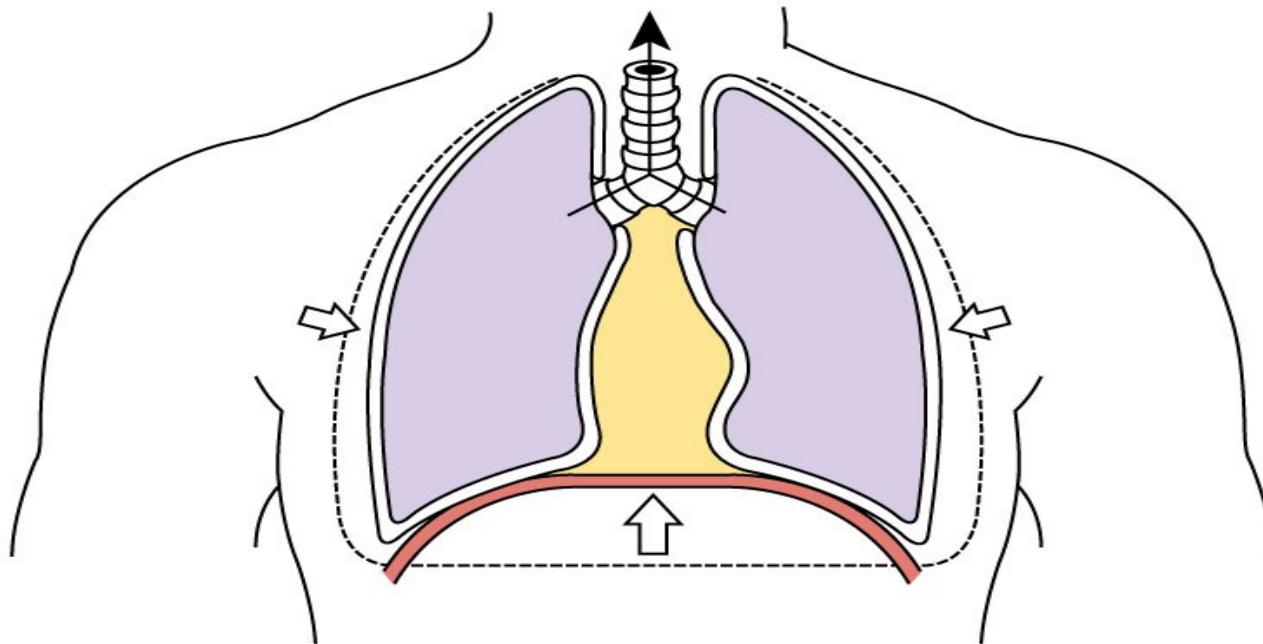
Diaphragm contracts, thoracic volume increases.

- also causing an increase in the transverse diameter of the thorax

Fig. 11 Diaphragm contraction

Expiration

- the vertical dimension of the chest cavity is decreased



- causing a decrease in the transverse

diameter of the thorax

Diaphragm relaxes, thoracic volume decreases.

Fig. 12 Diaphragm Relaxation

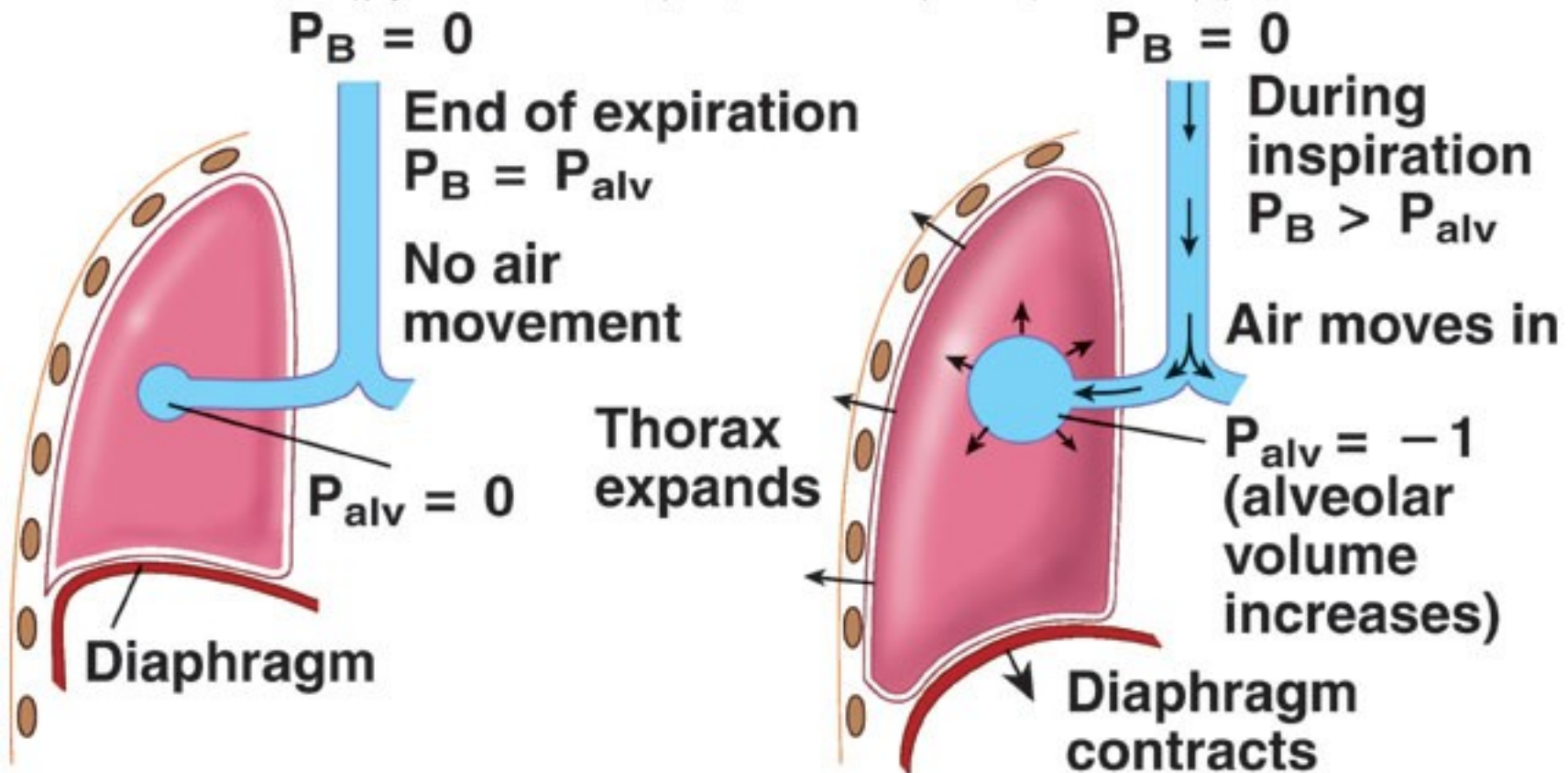
Inspiration/Inhalation

- Diaphragm & External intercostal muscles contract
- Increases volume in thoracic cavity as muscles contract
- Volume of lungs increases
- Intrapulmonary pressure decreases
- Alveolar pressure become sub-atmospheric pressure
- Air flows from atmosphere in to the lungs

- The second method for expanding the lungs is to raise the rib cage

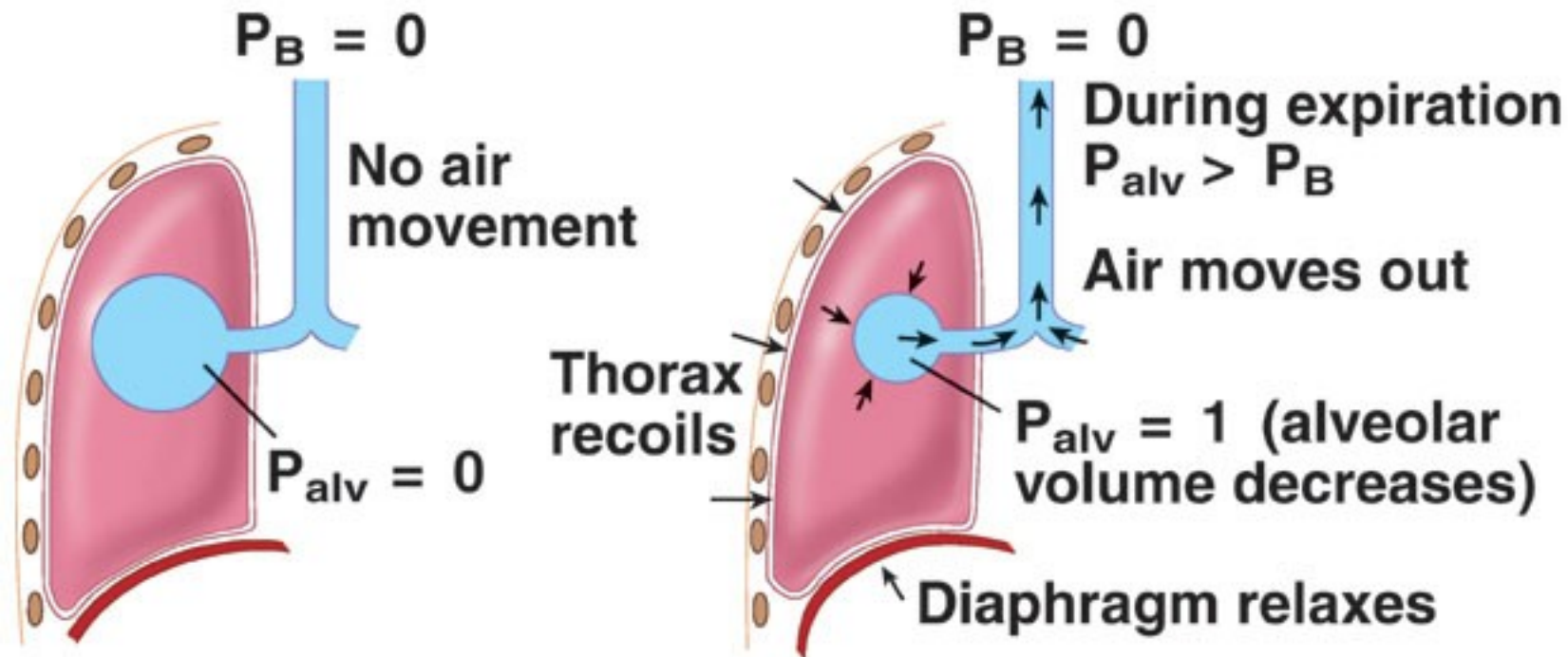
When **external intercostal muscles contract**, the ribs are pulled upward and forward, causing an increase in both the lateral and anteroposterior diameters

The **intercostal muscles** are supplied by **intercostal nerves**



1. Barometric air pressure (P_B) is equal to alveolar pressure (P_{alv}) and there is no air movement.
2. Increased thoracic volume results in increased alveolar volume and decreased alveolar pressure. Barometric air pressure is greater than alveolar pressure, and air moves into the lungs.

Fig. 16 Inspiration



3. End of inspiration.

4. Decreased thoracic volume results in decreased alveolar volume and increased alveolar pressure. Alveolar pressure is greater than barometric air pressure, and air moves out of the lungs.

Fig. 16 Expiration

Muscles of inspiration:

- diaphragm
- external intercostals muscles
- sternocleidomastoid muscles
- scalene muscles

Muscles of expiration (forced expiration):

- abdominal muscles (rectus abdominis, the external and internal oblique muscles)
- internal intercostals

Respiratory Muscles

Inspiration

Sternocleidomastoid
(elevates sternum)

Scalenes
(fix or elevate ribs 1–2)

External intercostals
(elevate ribs 2–12,
widen thoracic cavity)

Pectoralis minor
(elevates ribs 3–5)

Internal intercostals,
intercartilaginous part
(aid in elevating ribs)

Diaphragm
(descends and
increases depth
of thoracic cavity)

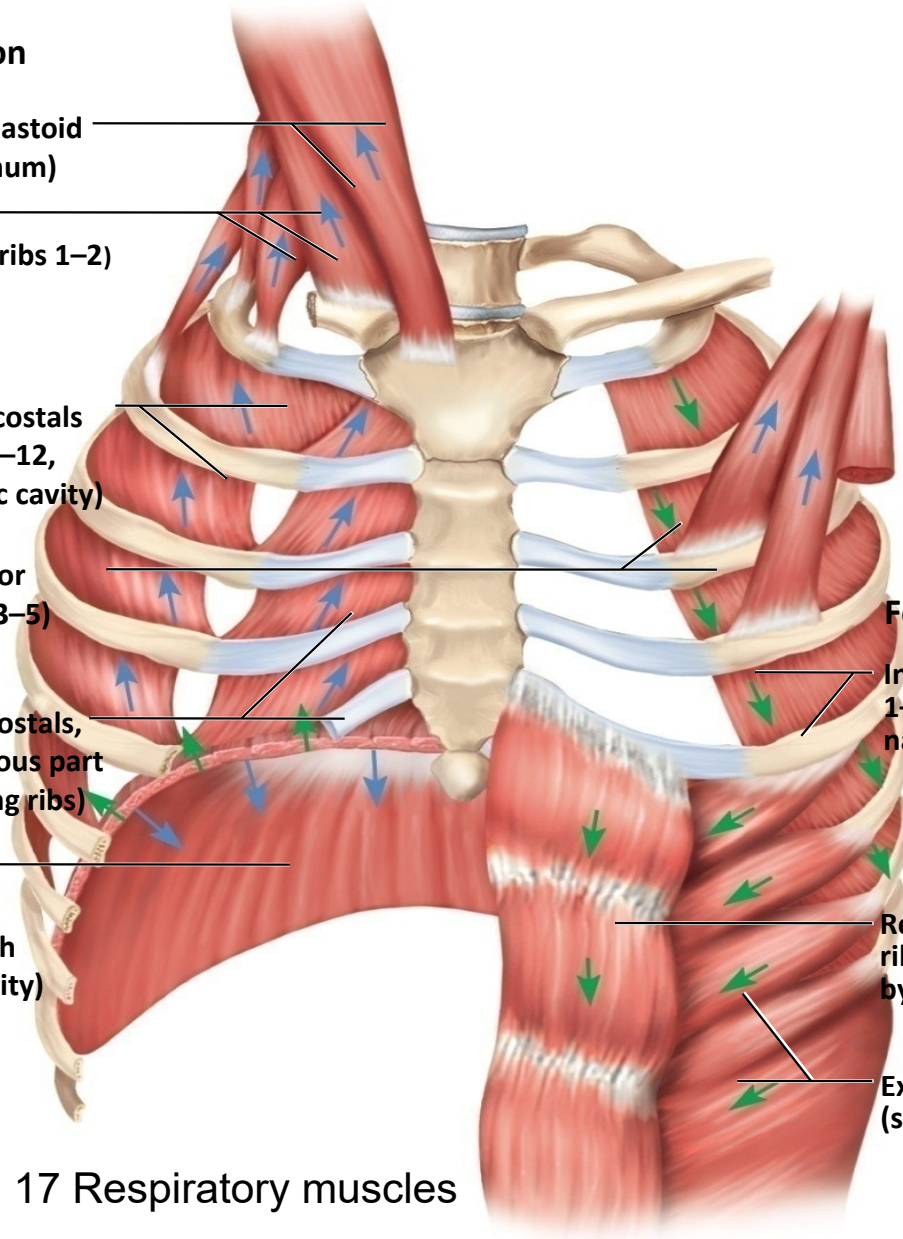
Forced expiration

Internal intercostals, (depress ribs
1–11,
narrow thoracic cavity)

Rectus abdominis (depresses lower
ribs, pushes diaphragm upward
by compressing abdominal organs)

External abdominal oblique
(same effects as rectus abdominis)

Fig. 17 Respiratory muscles



Pleural Pressure and Alveolar Pressure Change during Respiration

Pleural pressure (intrapleural pressure):

- is the pressure in the thin space between the lung pleura and the chest wall pleura
- a slightly *negative* pressure
- at the beginning of inspiration is about -5 centimeters of water (pressure)
- during inspiration intrapleural pressure become more negative, an average of about -7.5 centimeters of water

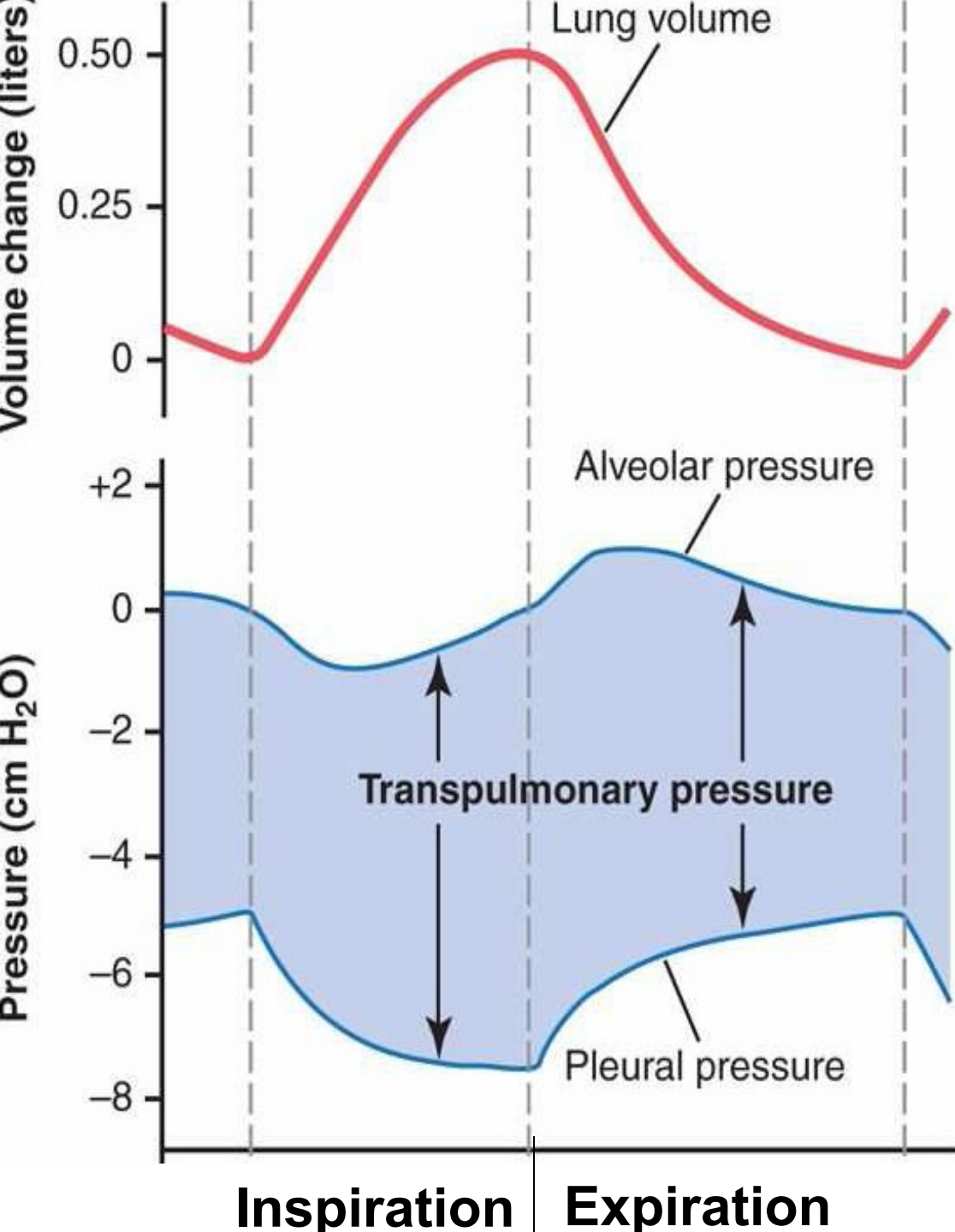


Fig. Alveolar, pleural and transpulmonary pressure

The elastic forces of the lungs:

- 1) Alveolar fluid surface tension
- 2) Elastic forces of the lung tissue

SURFACE TENSION

This is a force acting across an imaginary line 1 cm long in the surface of liquid lining alveoli and respiratory bronchioles.

Surface tension arises because of attractive forces b/n **adjacent liquid molecules on surface of alveoli**

This makes the alveolar surface area smaller (contracted) behaving like a soap bubble blown out of a tube and generates a pressure.

Pressure in alveoli keeps alveoli normally distended.

NB: The lung can be regarded as a collection of 300 million bubbles (alveoli)- inherently unstable structure.

- Because of the surface tension of liquid lining the alveoli, large forces develop that tend to **collapse the alveoli**.
- Fortunately, **surfactant** lower the surface tension dramatically and thus resist collapse tendency of the lung.
- Surfactant is a phospholipid and dipalmitoyl phosphatidyl choline, DPPC (lecithin- phospholipid and lipoprotein), mainly composed of lipoprotein.

The phospholipid component is responsible for reducing surface tension

NB: As shown in the figure on the next slide:

1. Large alveoli (large radii) have low pressure and are easy to keep open.
2. Small alveoli (small radii) have high pressure and are difficult to keep open. In the absence of a surfactant, the small alveoli have a tendency to collapse

Pressure = (2×surface tension)/radius of alveolus ($P = 2T/r$)

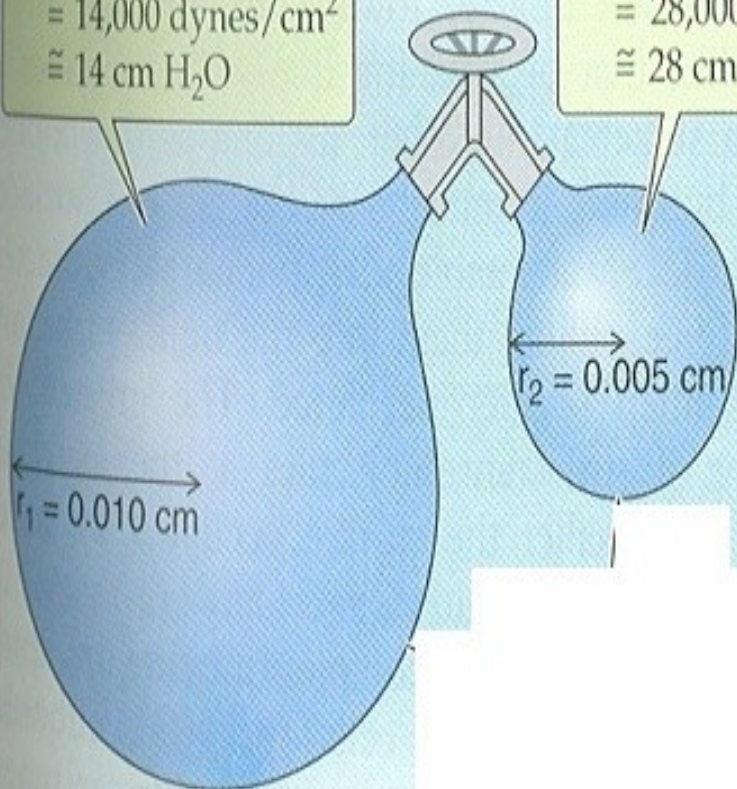
This is called the law of Laplace

Low pressure:

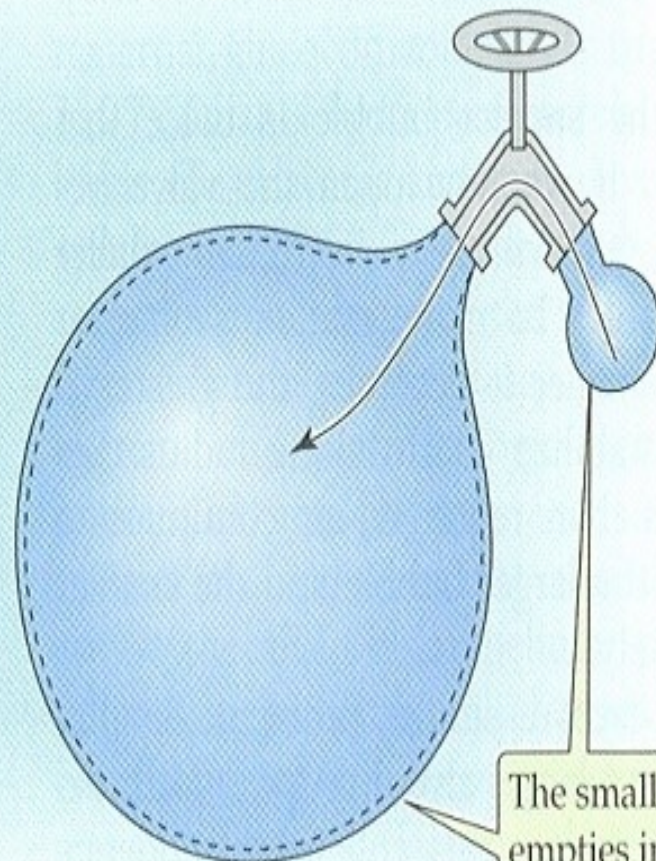
$$P_1 = \frac{2(70 \text{ dynes/cm})}{0.010 \text{ cm}}$$
$$= 14,000 \text{ dynes/cm}^2$$
$$\cong 14 \text{ cm H}_2\text{O}$$

High pressure:

$$P_2 = \frac{2(70 \text{ dynes/cm})}{0.005 \text{ cm}}$$
$$= 28,000 \text{ dynes/cm}^2$$
$$\cong 28 \text{ cm H}_2\text{O}$$



Remove
clamp



The small bubble
empties into the large
bubble.

Fig. Effect of Surface Tension on Lung

Effects of surfactant_

- Reduces surface tension → ↓ pressure necessary to inflate the lungs
→ ↓ *work of breathing*
- Reduces recoil force of lungs
- Increases distensibility of lungs
- Prevents alveolar collapse by balancing pressure btwn small and large alveoli

Respiratory distress syndrome of the newborn/ Infant respiratory distress syndrome (IRDS)/ neonatal respiratory distress syndrome: • is a syndrome in premature infants caused by developmental *insufficiency of surfactant production* and *structural immaturity* in the lungs (resulting primarily from abnormal surfactant)

- it can also result from a genetic problem with the production of surfactant associated proteins
- lungs have an extreme tendency to collapse

Compliance of the Lungs:

Lung compliance is *the extent to which the lungs will expand for each unit increase in transpulmonary pressure*

The total compliance of both lungs together in the normal adult human being *averages about 200 milliliters of air per centimeter of water*

Lung compliance are determined by are two types of the elastic forces of the lungs:

- (1) elastic forces of the lung tissue (represent $\frac{1}{3}$ of total lung elasticity)
- (2) elastic forces caused by surface tension of the fluid that lines the inside walls of the alveoli ($\frac{2}{3}$ of total lung elasticity)

A reduced lung compliance occurs in:

- an increase of fibrous tissue in the lung (pulmonary fibrosis)
- alveolar edema, which prevents the inflation of some alveoli
- when the pulmonary venous pressure is increased and the lung becomes engorged with blood

An increased lung compliance occurs when:

- pulmonary emphysema
- the normal aging lung

"Work" of Breathing

Under resting conditions, the respiratory muscles normally perform "work" to cause inspiration but not to cause expiration

The work of inspiration is divided into three fractions:

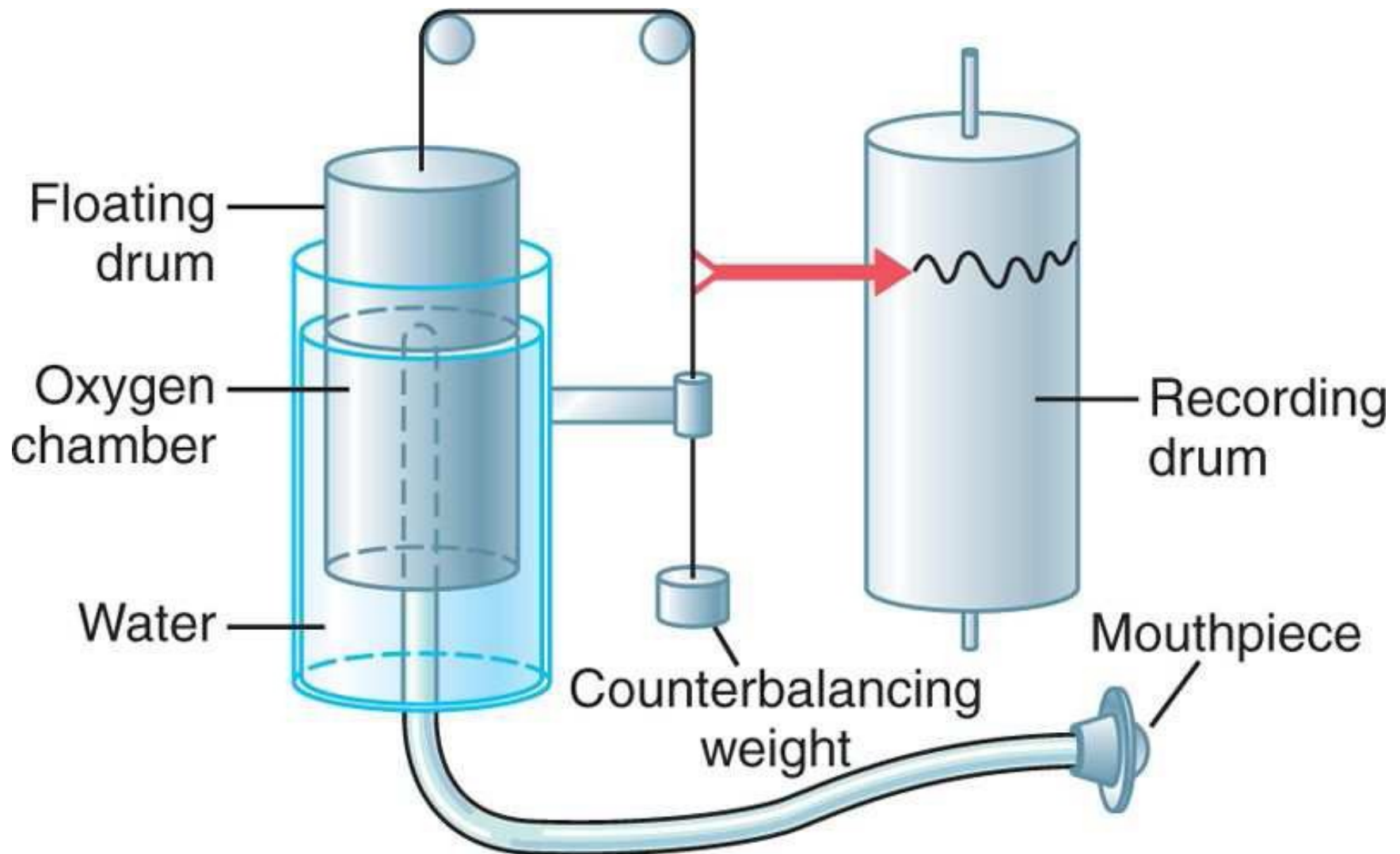
- (1) that required to expand the lungs against the lung and chest elastic forces, called *compliance work or elastic work*;
- (2) that required to overcome *the viscosity of the lung and chest wall structures*
- (3) that required to overcome airway resistance to movement of air into the lungs (*airway resistance work*)

During normal quiet respiration, only *3 to 5 percent* of the total energy expended by the body is required for pulmonary ventilation

Pulmonary Volumes and Capacities

Pulmonary ventilation can be studied by recording the volume movement of air into and out of the lungs, by a method called spirometry

A **spirometer** is a volume recorder instrument of volume movement of air into and out of the lungs



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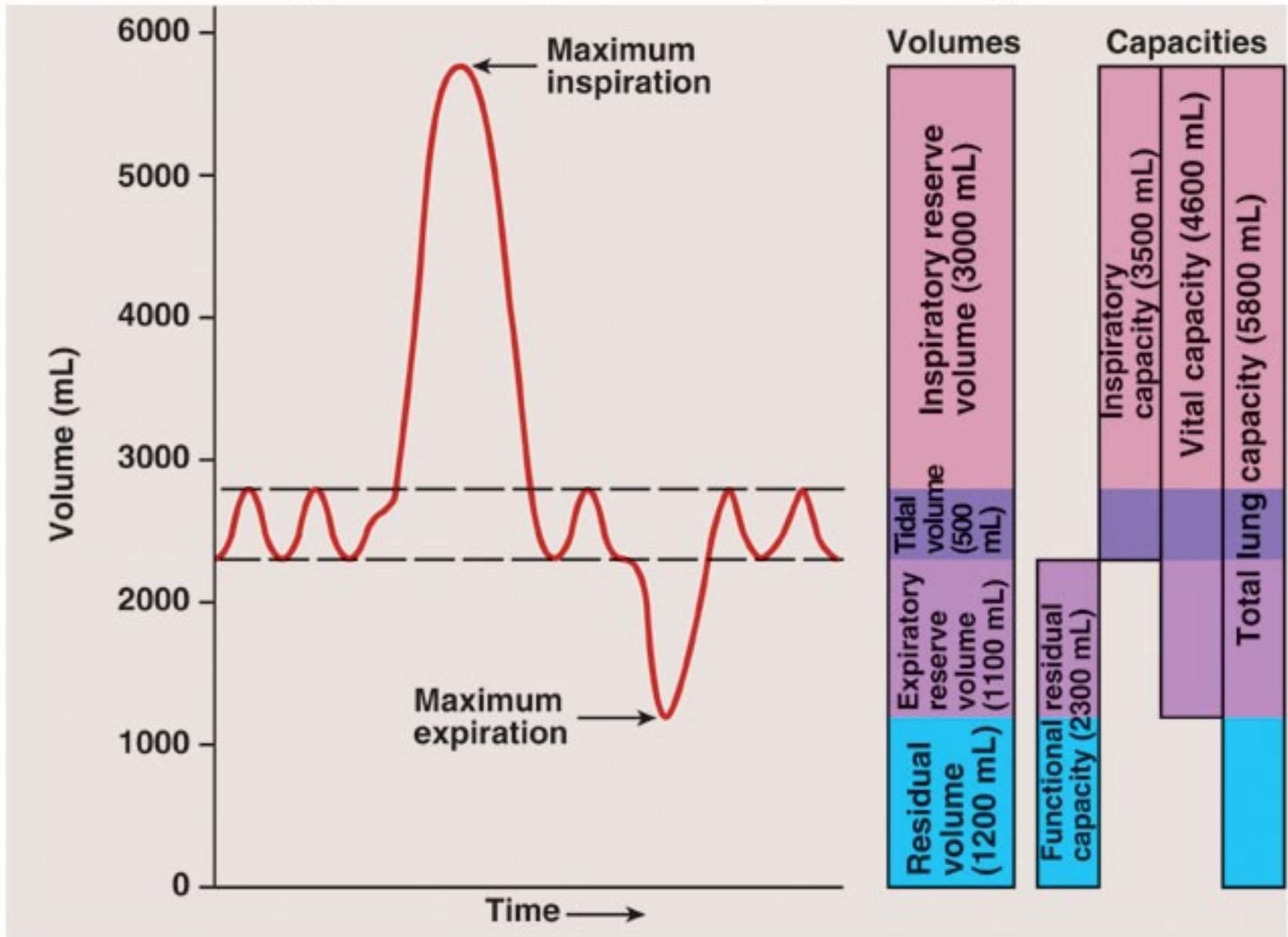
Figure Wet Spirometer

Pulmonary Volumes

There are four pulmonary volumes that, when added together, equal the maximum volume to which the lungs can be expanded

These volumes are:

- 1) *The tidal volume (V_T)*** is the volume of air inspired or expired with each normal breath; it amounts to about 500 milliliters in the adult male
- 2) *The inspiratory reserve volume (IRV)*** is the extra volume of air that can be inspired over and above the normal tidal volume when the person inspires with full force; it is usually equal to about 3000 milliliters (adult male)
- 3) *The expiratory reserve volume (ERV)*** is the maximum extra volume of air that can be expired by forceful expiration after the end of a normal tidal expiration; this normally amounts to about 1100 milliliters (adult male)
- 4) *The residual volume (RV)*** is the volume of air remaining in the lungs after the most forceful expiration; this volume averages about 1200 milliliters (adult male)



Pulmonary Capacities

Combinations of two or more of the volumes together are called *pulmonary capacities*

1) The inspiratory capacity (IC) equals the *tidal volume* plus the *inspiratory reserve volume*

- about 3500 milliliters (adult male)

2) The functional residual capacity (FRC) equals the *expiratory reserve volume* plus the *residual volume*

- about 2300 milliliters (adult male)

3) The vital capacity (VC) equals the *inspiratory reserve volume* plus the *tidal volume* plus the *expiratory reserve volume*

- about 4600 milliliters (adult male)

4) The total lung capacity (TLC) is the maximum volume to which the lungs can be expanded with the greatest possible effort (about 5800 milliliters, *for adult male*); it is equal to the *vital capacity* plus the *residual volume*

NB

All the values for lung volumes and lung capacities given above are the average for a *young adult man*

All pulmonary volumes and capacities are about 20 to 25 percent less in women than in men, and they are greater (larger) in large and athletic people than in small and non-athletic people

Pulmonary capacities

$$VC = IRV + V_T + ERV$$

$$VC = IC + ERV$$

$$TLC = VC + RV$$

$$TLC = IC + FRC$$

$$FRC = ERV + RV$$

Not all lung volumes & capacities can be measured directly by spirometry

Because the lungs cannot be emptied completely following forced expiration, **RV**, **FRC** and **TLC** cannot be measured directly by simple spirometry

Thus they are measured indirectly using a **dilution technique** using helium by the following procedure:

- the subject is connected to a spirometer
- filled with 10% helium in oxygen
(the lungs initially contain no helium)

before breathing from the spirometer, the person expires normally

- at the end of this expiration, the remaining volume in the lungs is equal to the functional residual capacity (**FRC**)
 - at this point, the subject immediately begins to breathe from the spirometer, and the gases of the spirometer mix with the gases of the lungs

- after the subject rebreathes the helium and oxygen mixture and equilibrates with the spirometer, the helium concentration in the lungs will become the same as in the spirometer

From the conservation of mass principle, we can write:

$$C_1 \times V_1 = C_2 \times (V_1 + V_2)$$

$$V_2 = (V_1 \times (C_1 - C_2)) \div C_2$$

$$C_1 \times V_1 = C_2 \times (V_1 + V_2)$$

$$V_2 = (V_1 \times (C_1 - C_2)) \div C_2$$

Where: C_1 = initial concentration of helium in the spirometer

V_1 = the initial volume of helium and oxygen mixture
in the spirometer

C_2 = helium concentration after equilibration

V_2 = unknown volume in the lungs (functional residual
capacity = FRC)

Once the FRC has been determined RV and the total lung capacity TLC can be determined as follows:

$$RV = FRC - ERV$$

$$TLC = FRC + IC$$

Another way of measuring the functional residual capacity (FRC) is with a **body plethysmograph**

Minute Ventilation (Minute Respiratory Volume)

The *minute respiratory volume*/minute ventilation:

- is the total amount of new air moved into the respiratory passages each minute
- this is equal to the *tidal volume* X the *respiratory rate per minute*

Breathing frequency (resting)

Measured by :

Observing chest movement

Stetography

= 12 breaths /min in adults

=14-16 breaths /min in newborns

Normal tidal volume is about 500 milliliters (adult)

Therefore, the *minute respiratory volume averages about 6 L/min (adult)*

Alveolar ventilation:

is the rate at which new air reaches the gas exchange areas of the lungs (alveoli, alveolar sacs, alveolar ducts, and respiratory bronchioles)

some of the air a person breathes never reaches the gas exchange areas but simply fills respiratory passages where gas exchange does not occur (the conducting zone), such as the nose, pharynx, and trachea

This space (conducting zone) is called the *anatomic dead space*

The normal anatomical dead space air in a young adult man is about 150 milliliters

On occasion, some of the alveoli themselves are non-functional or only partially functional (gas exchange) because of absent or poor blood flow through the adjacent pulmonary capillaries

Therefore, from a functional point of view, these alveoli must also be considered dead space (forms alveolar dead space)

When the alveolar dead space is included in the total measurement of dead space, this is called the *physiologic dead space*

Alveolar ventilation per minute:

- is the total volume of new air entering the alveoli and adjacent gas exchange areas each minute

$$V_A = \text{Freq.} \times (V_T - V_D)$$

where;

V_A = the volume of alveolar ventilation per minute

Freq = the frequency of breathing per minute

V_T = the tidal volume

V_D = the physiologic dead space volume

In an adult male;

$$V_A = (500\text{ml} - 150\text{ml}) \times 12 = 4200 \text{ ml/min}$$

Alveolar ventilation can be increased by raising either tidal volume or respiratory frequency (or both)

↑ v_T is often more effective because this reduces the proportion of each breath occupied by the anatomic dead space

Pulmonary Circulation

The lung has two circulations:

(1) Pulmonary circulation (2) Bronchial circulation

1) Pulmonary Circulation (A low-pressure, high-flow circulation)

- this is the circulation that supplies venous blood from all parts of the body to the **alveolar capillaries** where oxygen is added and carbon dioxide is removed from the blood

- pulmonary artery and its arterial branches carry blood to the alveolar capillaries for gas exchange and the pulmonary veins then return the blood to the left atrium

The pulmonary artery:

- is thin with a wall thickness one third that of the aorta
- pulmonary arterial branches are very short
- all the pulmonary arteries, even the smaller arteries and arterioles, have larger diameters than their counterpart systemic arteries
- the pulmonary veins, like the pulmonary arteries, are also short so, they immediately empty their effluent blood into the left atrium

2) **Bronchial circulation (A high-pressure, low-flow circulation)**

- it is part of the systemic circulation
- this circulation supplies systemic arterial blood to:
 - ✓ the trachea, the bronchial tree including the terminal bronchioles
 - ✓ the supporting tissues of the lung, and
 - ✓ the outer coats (adventitia) of the pulmonary arteries and veins

The bronchial arterial blood is *oxygen-rich* blood, in contrast to the oxygen-poor blood in the pulmonary arteries

The bronchial circulation does *not extend to the respiratory zone*, which is served by the pulmonary circulation

Blood flow through the bronchial circulation is only *1–2% of cardiac output*

Bronchial veins from large airways return about half the bronchial blood flow to the right heart via the Azygus vein

The other half of the bronchial circulation drains directly into the pulmonary circulation (adds oxygen-poor blood to the oxygen-rich blood returning to the left heart)

Blood Flow Through the Lungs and Its Distribution

Blood volume of the lungs is about 450 milliliters, (about 9% of the total blood volume of the entire circulatory system)

- 70 milliliters of this pulmonary blood volume is in the pulmonary capillaries
- the remainder is divided about equally between the pulmonary arteries and the veins

Yet the total blood flow through the lungs each minute is the same as through the systemic circulation; the blood flow through the lungs is essentially equal to the cardiac output

Effect of oxygen level (concentration) in alveoli on pulmonary blood flow

For adequate aeration of the blood to occur, it is important for the blood to be distributed to those segments of the lungs where the alveoli are best oxygenated

Decreased *alveolar oxygen* reduces local alveolar blood flow and *regulates pulmonary blood flow distribution*

At concentration level of oxygen in the air of the alveoli below 70 % of normal (below 73 mm Hg PO₂), the adjacent blood vessels constrict (opposite to the effect observed in systemic vessels)

This causes the blood to flow through other areas of the lungs that are better ventilated

Effect of Hydrostatic Pressure Gradients in the Lungs on Regional Pulmonary Blood Flow

It is known that the blood pressure in the foot of a standing person can be as much as 90 mm Hg greater than the pressure at the level of the heart

The same trend but to a lesser degree, occurs in the lungs

The pulmonary arterial pressure in the uppermost portion of the lung of a standing person is about 15 mm Hg less than the pulmonary arterial pressure at the level of the heart, and the pressure in the lowest portion of the lungs is about 8 mm Hg greater

So there is **little blood flow in the top (apex) of the lung** but about **five times as much flow in the bottom (base) of the lung**

The three possible zones (patterns) of pulmonary blood flow:

Zone 1: *No blood flow during all portions of the cardiac cycle* because the local alveolar capillary pressure in that area of the lung never rises higher than the alveolar air pressure during any part of the cardiac cycle

The pressure gradient in zone 1 is represented as

$$PA > Pa > Pv$$

where

PA = Alveolar pressure

Pa = Pulmonary artery pressure

Pv = pulmonary venous pressure

Zone 1 is usually *small or nonexistent* in healthy people because the pulsatile pulmonary arterial pressure is sufficient to keep the capillaries partially open at the apex

When zone 1 does occur; it occurs in the apex of the lungs
Zone 1 may easily be created by conditions that elevate alveolar pressure or decrease pulmonary arterial pressure

Zone 2: *Intermittent blood flow* only during the peaks of pulmonary arterial pressure because the systolic pressure is then greater than the alveolar air pressure, but the diastolic pressure is less than the alveolar air pressure

The pressure gradient in zone 2 is represented as
 $P_a > P_A > P_v$

Zone 2 blood flow condition occurs in apical part of the lungs

Blood flow in zone 2 depends on the difference between arterial and alveolar pressures but not by the arterial-venous pressure difference

Zone 3: Continuous blood flow because the alveolar capillary pressure remains greater than alveolar air pressure during the entire cardiac cycle

The pressure gradient in zone 3 is represented as

$$P_a > P_v > P_A$$

This occurs in the *lower regions of the lungs*, from about 10 cm above the level of the heart all the way to the bottom of the lungs

In zone 3 blood flow is determined by the usual arterial-venous pressure difference

Note that

Normally, the lungs have only zones 2 and 3 blood flow, zone 2 (intermittent flow) in the apices and zone 3 (continuous flow) in all the lower areas

In an upright person, pulmonary blood flow decreases almost linearly from the base to apex

Blood flow distribution is affected by gravity and can be altered by changes in body positions

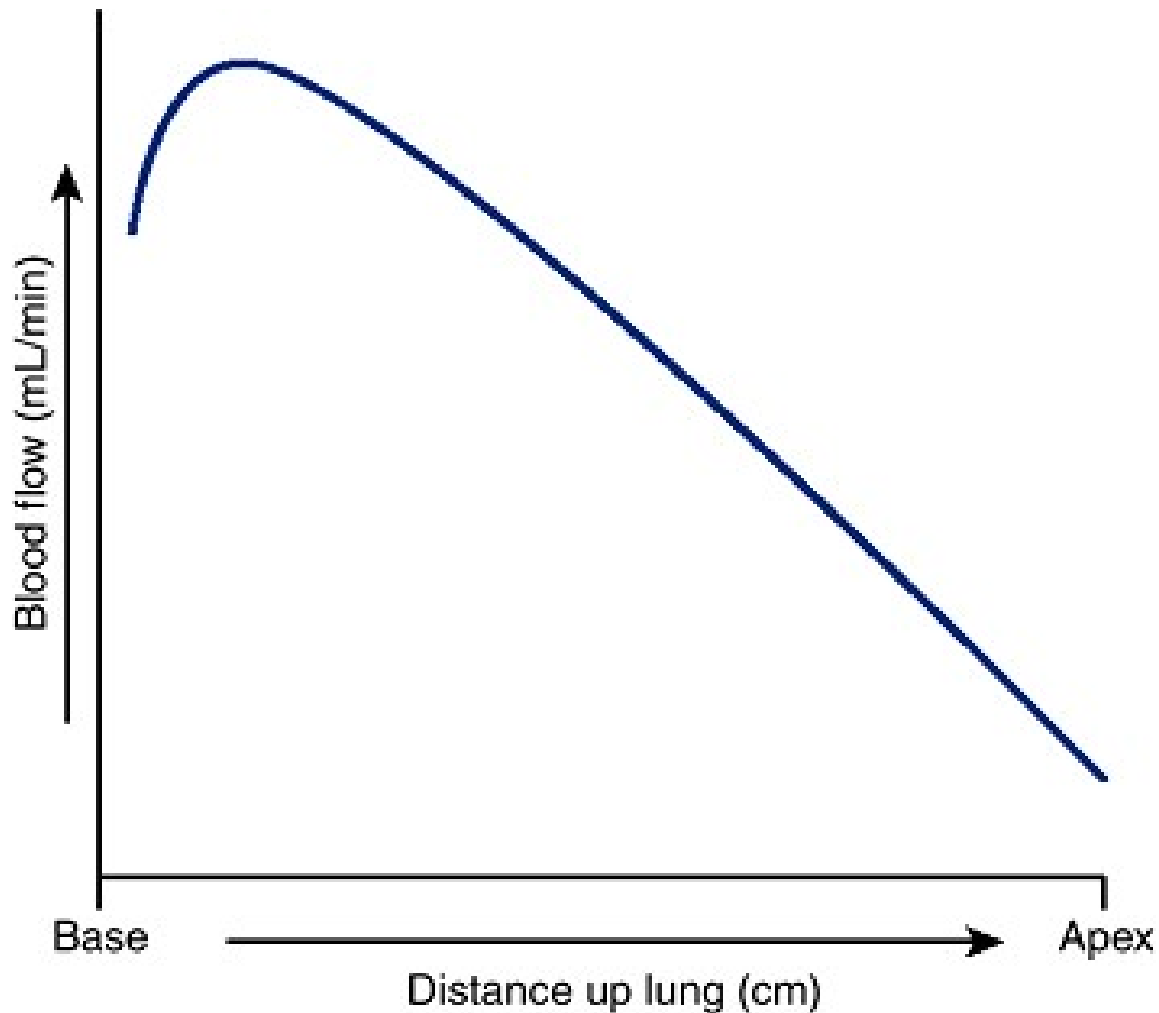


Figure Gravity causes uneven pulmonary blood flow in the upright person. Pulmonary blood flow is low at the apex and toward the base of the lungs there is an increase in blood flow

Exercise tends to offset the gravitational effects on blood flow in an upright person

Cardiac output increases during exercise, that causes increased pulmonary arterial pressure

Increased pulmonary artery pressure leads to capillary recruitment and distention in the lungs apex

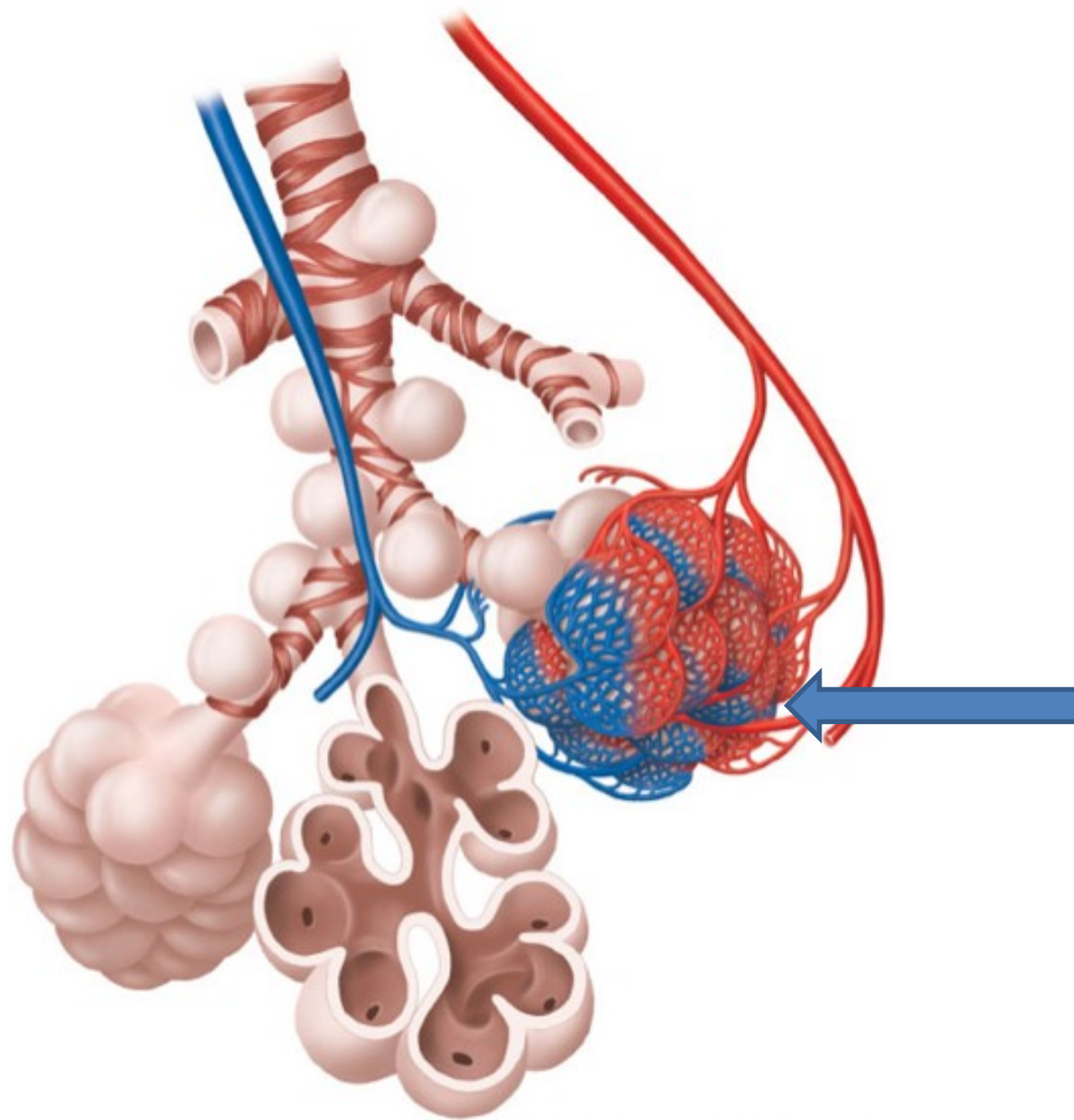
These resulting in increased blood flow and minimizing regional differences in blood flow in the lungs

Pulmonary Capillary Dynamics

Alveolar walls are lined with so many capillaries

In most places, the capillaries almost touch one another side by side thus, it is often said that the capillary blood flows in the alveolar walls as a "**sheet of flow**," rather than in individual capillaries

When the cardiac output is normal, blood passes through the pulmonary capillaries in about 0.8 second



**Pulmonary
Capillaries**

Capillary Exchange of Fluid in the Lungs and Pulmonary Interstitial Fluid Dynamics

The dynamics of fluid exchange across the lung capillary membranes are *qualitatively* the same as for peripheral tissues

However, *quantitatively*, there are important differences, as follows:

- 1) *The pulmonary capillary pressure is about 7 mm Hg (favors filtration)*

- 2) *The interstitial fluid pressure (-8 mm Hg) in the lung*
- interstitial fluid is slightly more negative than that in the peripheral subcutaneous tissue (favors filtration)
- 3) The pulmonary capillaries are relatively leaky to protein molecules, so *the colloid osmotic pressure of the pulmonary interstitial fluid is about 14 mm Hg*, in comparison with less than half this value in the peripheral tissues (8 mm Hg) (favors filtration)
- 4) *Plasma colloid osmotic pressure is about 28 mm Hg* (opposes filtration)

Forces that cause fluid movement across pulmonary capillaries	mm Hg
<i>Forces tending to cause movement of fluid outward from the capillaries and into the pulmonary interstitium:</i>	
Capillary pressure	7
Interstitial fluid colloid osmotic pressure	14
Negative interstitial fluid pressure	<u>8</u>
TOTAL OUTWARD FORCE	29
<i>Forces tending to cause absorption of fluid into the capillaries:</i>	
Plasma colloid osmotic pressure	<u>28</u>
TOTAL INWARD FORCE	28
Net filtration force	29 – 28 = 1

The normal outward forces are slightly greater than the inward forces

This causes a slight continual flow of fluid from the pulmonary capillaries into the lung interstitial spaces

This fluid is pumped back to the circulation through the pulmonary lymphatic system

A small amount of the fluid also evaporates in the alveoli

Pulmonary Edema

Filling of the pulmonary interstitial spaces and alveoli with large amounts of free fluid

Results from:

- increase in fluid filtration out of the pulmonary capillaries (above lymphatic function)
- impediment in pulmonary lymphatic function

The most common causes of pulmonary edema are:

- left-sided heart failure or mitral valve disease, with consequent great increases in pulmonary venous pressure and pulmonary capillary pressure
- damage to the pulmonary blood capillary membranes caused by infections or by breathing noxious substances such as chlorine gas or sulfur dioxide gas (cause leakage of protein and fluid out of pulmonary capillaries)

Animal experiments indicated that the pulmonary capillary pressure normally must rise to a value at least equal to the colloid osmotic pressure of the plasma inside the capillaries before significant pulmonary edema will occur

Thus, in the human being we can predict that the pulmonary capillary pressure must rise from the normal level of 7 mm Hg to more than 28 mm Hg to cause pulmonary edema, giving an *acute safety factor against pulmonary edema* of 21 mm Hg

Gas Exchange Through the Respiratory Membrane

After alveoli are ventilated with fresh air then *diffusion* of oxygen from the alveoli into the pulmonary blood and diffusion of carbon dioxide in the opposite direction out of the blood occurs due to partial pressure difference between alveoli and pulmonary capillary blood for the gases

Gas Pressures in a Mixture of Gases "Partial Pressures"

Atmospheric air is mixture of gases, mainly of *nitrogen*, *oxygen*, and *carbon dioxide*

The total pressure of this mixture at sea level averages 760 mm Hg
The pressure caused by particular gas alone is called the *partial pressure of that gas*

Each gas in the mixture contributes to the total pressure in direct proportion to its concentration

Example

Atmospheric Air (mm Hg)		Partial pressure of the gas mm Hg	
N ₂	(78.62%)	0.786×760 =	597.0
O ₂	(20.84%)	0.208×760 =	159.0
CO ₂	(0.04%)		0.3
H ₂ O	(0.50%)		3.7
TOTAL	(100.0%)		760.0

The partial pressures of individual gases in a mixture are designated by the symbols as follows:

PO₂ = partial pressure of Oxygen

PCO₂ = partial pressure of Carbon dioxide

Gases dissolved in water or in body tissues fluid also exert pressure because the dissolved gas molecules are moving randomly and have kinetic energy

The partial pressure of a gas in a solution is determined not only by its concentration but also by the *solubility coefficient* of the gas

These relations are expressed by the following formula, which is *Henry's law*:

Partial pressure $\propto$ $\frac{\text{concentration of dissolved gases}}{\text{solubility coefficient}}$

** The volume of a gas that can be dissolved by a unit volume of solvent at a specified pressure and temperature is called solubility coefficient.

Carbon dioxide is more than 20 times as soluble as oxygen

So, for a given concentration, the partial pressure of carbon dioxide is less than one-twentieth that of oxygen

Diffusion of Gases Between the Alveoli and the Pulmonary Blood (through respiratory membrane)

Exchange of gases between **the alveolar air and the pulmonary blood** occurs through the membranes of the terminal portions of the respiratory passage termed as **the respiratory zone** (**respiratory bronchioles, alveolar duct and alveoli**)

This membrane that allows gas exchange in the lung is termed as ***respiratory membrane (pulmonary membrane)***

The respiratory membrane:

- **exceedingly thin** (averages about **0.6 micrometer**, in some areas is as little as **0.2 micrometer**)
- **has a surface area** of between 50 and 100 square meters (av. **70 square meters**)

So, well suited to its function of gas exchange

Layers of the respiratory membrane wall are:

- 1) A layer of fluid lining the alveolus and containing surfactant
- 2) The alveolar epithelium composed of thin epithelial cells
- 3) An epithelial basement membrane
- 4) A thin interstitial space between the alveolar epithelium basement membrane and the capillary basement membrane
- 5) A capillary basement membrane that in many places fuses with the alveolar epithelial basement membrane
- 6) The capillary endothelial membrane

Diffusion of gases b/n the alveoli and the pulmonary blood

The partial pressure of each gas in the alveolar respiratory gases mixture **tends to force molecules of that gas into solution in the blood** of the alveolar capillaries

Conversely, the molecules of the same gas that are already dissolved in the blood are bouncing randomly in the fluid of the blood, and some of these bouncing **molecules escape back into the alveoli**

The rate of diffusion across the respiratory membrane is directly proportional to the partial pressure difference across the membrane the gas

Thus, **net diffusion is determined by the difference between the two partial pressures (partial pressure of the gas in alveoli and in the capillary blood)**

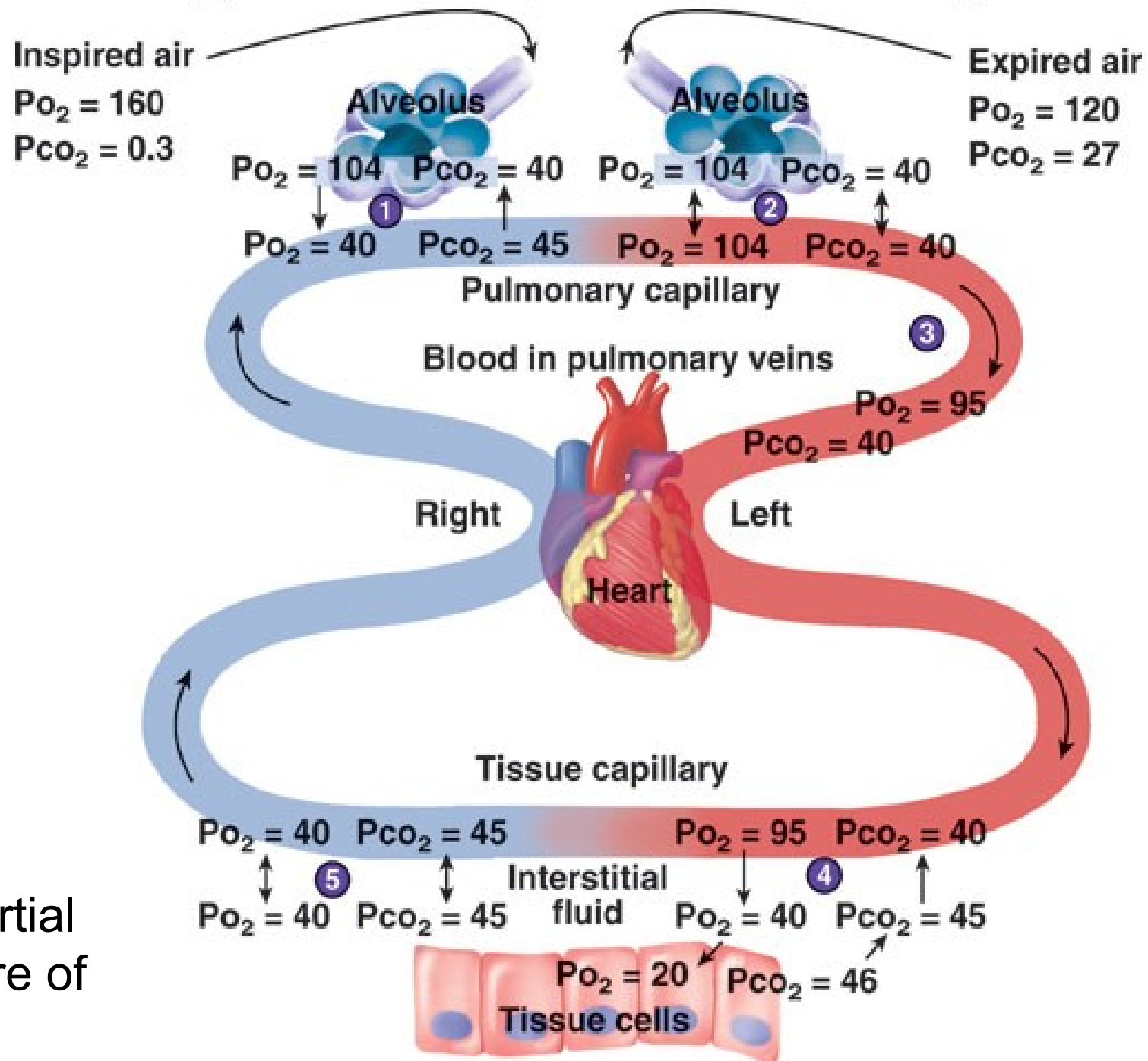


Fig. Partial pressure of gases

Oxygen diffuses (net) from alveoli into the blood of pulmonary capillary

For carbon dioxide net diffusion occurs toward the gas phase in the alveoli from pulmonary capillary blood

Alveolar air does not have the same concentrations of gases as atmospheric air because of:

- 1) the alveolar air is only partially replaced by atmospheric air with each breath
- 2) oxygen is constantly being absorbed into the pulmonary blood from the alveolar air
- 3) carbon dioxide is constantly diffusing from the pulmonary blood into the alveoli
- 4) dry atmospheric air that enters the respiratory passages is humidified even before it reaches the alveoli

Factors affecting the rate of gas diffusion across respiratory membrane:

- 1) solubility of the gas in the membrane
- 2) the cross-sectional area of the membrane
the distance through which the gas must diffuse (thickness of the membrane)
- 4) the molecular weight of the gas, and
- 5) the temperature of the membrane

Ventilation-Perfusion ratio

The ventilation-perfusion ratio is expressed as VA/Q where

V_A = alveolar ventilation

Q = alveolar capillary blood flow

When the ventilation (VA) is zero, yet there is still perfusion (Q) of the alveolus, the VA/Q is zero

When there is adequate ventilation (VA) but zero perfusion (Q), the ratio VA/Q is infinity

At a ventilation-perfusion ratio of either zero or infinity, there is no exchange of gases through the respiratory membrane of the affected alveoli

When V_A/Q is equal to zero air in the alveolus comes to equilibrium with the venous blood oxygen and carbon dioxide (PO_2 of 40 mm Hg and a PCO_2 of 45 mm Hg);

✓ so, there is no gas exchange because of absence of partial pressure difference across respiratory membrane

When V_A/Q equals infinity the alveolar air becomes equal to the humidified inspired air (PO_2 of 149 mm Hg and a PCO_2 of 0.3 mm Hg);

✓ no gas because of absence of capillary blood flow to carry oxygen away or to bring carbon dioxide to the alveoli

When there is both normal alveolar ventilation and normal alveolar capillary blood flow (normal alveolar perfusion), exchange of oxygen and carbon dioxide through the respiratory membrane is nearly optimal, (alveolar PO_2 is 104 mm Hg, alveolar PCO_2 is 40 mm Hg)

Whenever VA/Q is below normal, there is inadequate ventilation to provide the oxygen needed to fully oxygenate the blood flowing through the alveolar capillaries

Therefore, a certain fraction of the venous blood passing through the pulmonary capillaries does not become oxygenated

This fraction is called *shunted blood*

Also, some additional **blood flows through bronchial vessels** rather than through alveolar capillaries, normally about 2 percent of the cardiac output; this, too, is unoxygenated, **shunted blood**

The total quantitative amount of shunted blood per minute is called ***the physiologic shunt***

The greater the physiologic shunt, the greater the *amount of blood that fails to be oxygenated* as it passes through the lungs

When **ventilation of some of the alveoli is great** but **alveolar blood flow is low**, there is far more available oxygen in the alveoli than can be transported away from the alveoli by the flowing blood

Thus, the ventilation of these alveoli is said to be ***wasted***

The ventilation of the anatomical dead space areas of the respiratory passageways is also wasted

Abnormal VA/Q in the Upper and Lower Parts of Normal Lung

In a normal person in the upright position, both pulmonary capillary blood flow and alveolar ventilation are considerably less in the upper part of the lung than in the lower part

Toward lung apex blood flow is decreased considerably more than ventilation is

Ventilation increases slowly from top to bottom of the lung and blood flow increases more rapidly

Thus, the ventilation-perfusion ratio is abnormally high at the apex of the lung (where the blood flow is minimal)

At the top (apex) of the lung:

- VA/Q ratio is as much as 2.5 times as great as the ideal value
- There is a moderate degree of *dead space* in this area of the lung

In the bottom (base) of the lung:

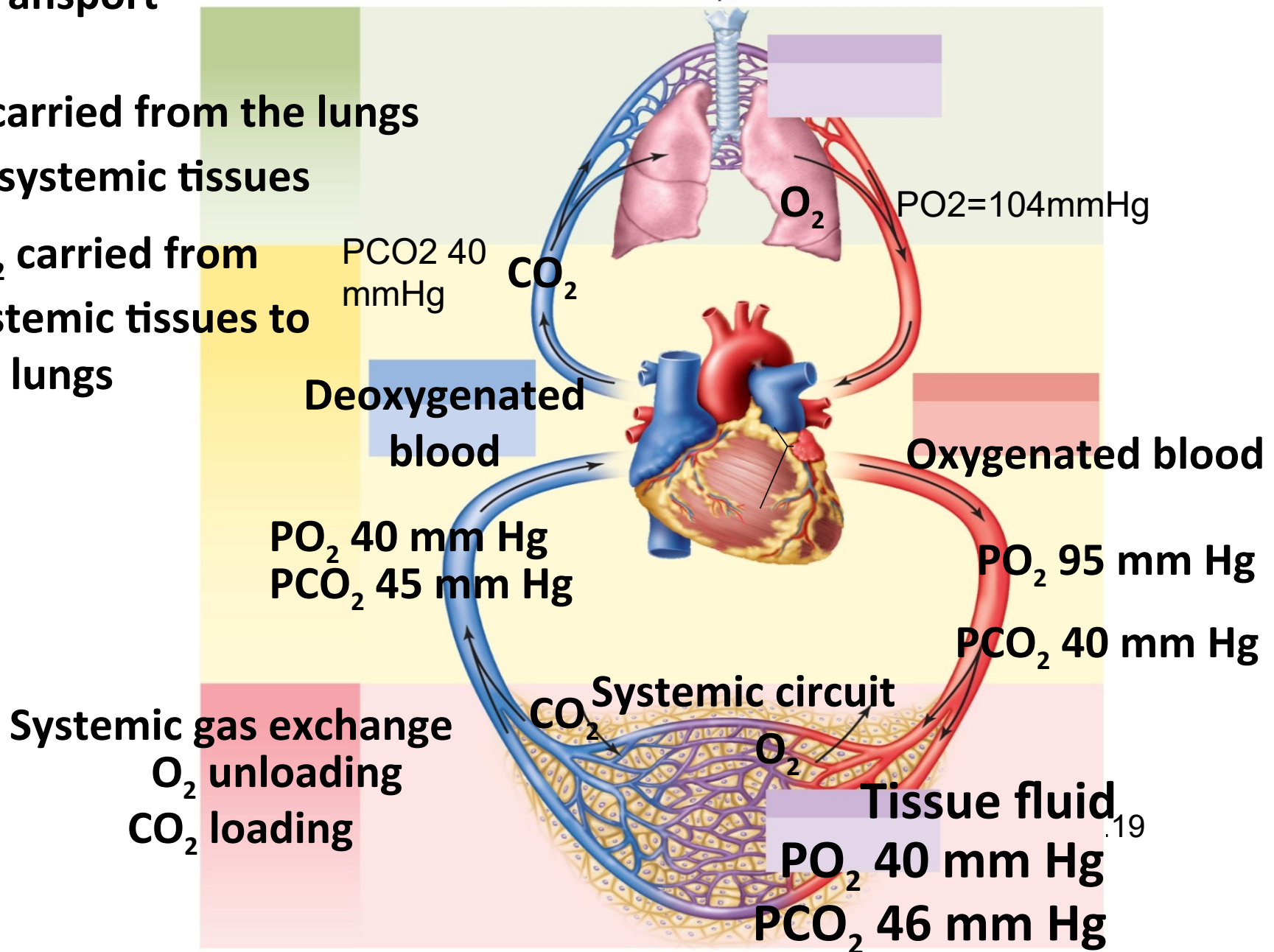
- There is slightly too little ventilation in relation to blood flow
- VA/Q as low as 0.6 times the ideal value
- Small fraction of the blood fails to become normally oxygenated, and there is *shunted blood*

4) Transport of Oxygen and Carbon Dioxide in Blood

Transport of oxygen and carbon dioxide by the blood depends on both *diffusion of gases and the flow of blood*

Gas transport

- O_2 carried from the lungs to systemic tissues
- CO_2 carried from systemic tissues to the lungs



Transport of Oxygen

Oxygen is transported by blood from lungs to tissues in two forms:

- 1) combined with Hb in the red cells (97 %)=Chemical form
- 2) dissolved in the blood (3 %) =physical form

Transport of Oxygen in the Dissolved State

At PO_2 of 95 mm Hg, about 0.29 milliliter of oxygen is dissolved in every 100 milliliters of water in the blood (arterial blood)

When the PO_2 of the blood falls to the normal 40 mm Hg in the tissue capillaries only 0.12 milliliters of oxygen remains dissolved in 100 milliliters of water in the blood (venous blood)

Thus, 0.17 milliliters of oxygen is normally transported in the dissolved state to the tissues by each 100 milliliters of arterial blood flow

Transport of Oxygen by Hemoglobin

Hemoglobin:

- Is the protein molecule contained in the erythrocytes
- Is important for transport of molecular oxygen (O_2) from the lungs into the tissues, and carbon dioxide (CO_2) from the tissues back into the lungs
- It is the molecular form of O_2 that is transported by Hb, not the ionic form to facilitate easy unbinding at the tissue level.
- Hemoglobin is also important for pH regulation

Each gram of hemoglobin can bind with 1.34 ml of oxygen (1.39 ml when the hemoglobin is chemically pure, but impurities such as methemoglobin reduce this)

Oxygen-carrying capacity (maximum amount of oxygen that can be carried by hemoglobin) is about 20 ml O₂/dL blood in a healthy adult male

Oxygen content is the amount of oxygen actually bound to hemoglobin (whereas capacity is the amount that can potentially be bound)

The **percentage saturation** of hemoglobin with oxygen (SO_2 %) is calculated from the ratio of oxyhemoglobin content over capacity:

$$SO_2 \% = \frac{HbO_2 \text{ Content}}{HbO_2 \text{ Capacity}} \times 100\%$$

For example, if oxygen content is 16 ml O_2 /dL blood and oxygen capacity is 20 ml O_2 /dL blood,

$$SO_2 \% = 80\%$$

The O_2 saturation of Hb is the percentage of the available binding sites that have O_2 attached

The percent O_2 saturation of arterial blood with PO_2 of 100 mm Hg is about 97%, whereas that of mixed venous blood with a PO_2 of 40 mm Hg is about 75%

Amount O₂ transported by hemoglobin

Under normal condition:

O₂ bound with Hb in normal systemic arterial blood is about 19.4 ml per 100 ml of blood (which is 97% saturated). On passing through the tissue capillaries, this amount is reduced, on average to 14.4 ml (PO₂ of 40 mm Hg, 75% saturated Hb). thus, *about 5 ml of oxygen is transported from the lungs to the tissues by each 100 ml of blood flow by means of Hb*

During heavy exercise:

- Muscle cells use oxygen at a rapid rate
- In extreme cases, exercise can cause the muscle interstitial fluid PO₂ to fall from the normal 40 mm Hg to as low as 15 mm Hg
- At this low pressure, only 4.4 ml of O₂ remain bound with the **Hb** in

each 100 milliliters of blood

Thus, 19.4 - 4.4 = **15 ml**, is the quantity of O₂ actually delivered to the tissues by each 100 milliliters of blood flow

Oxygen-Hemoglobin Dissociation/Association Curve

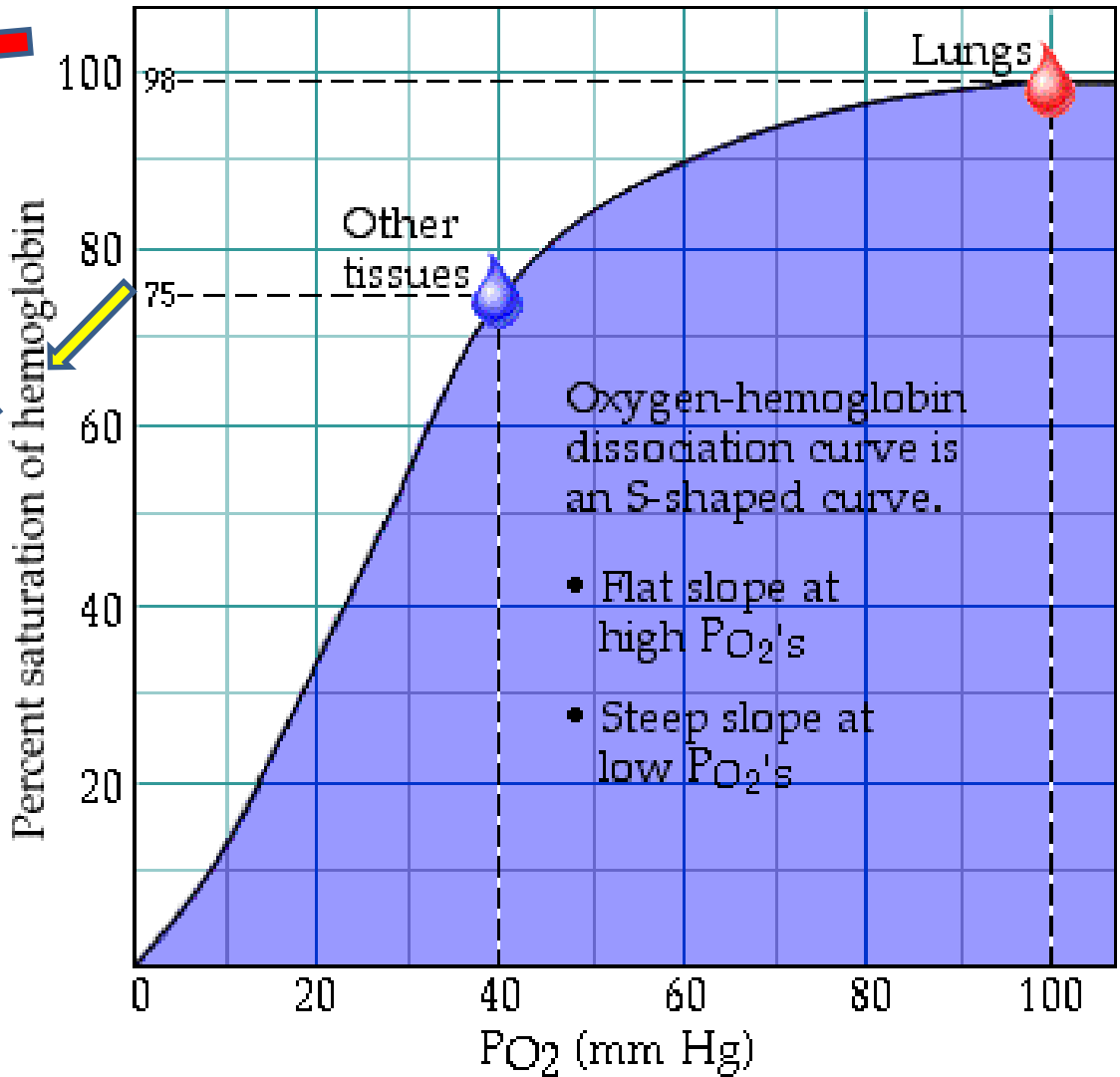
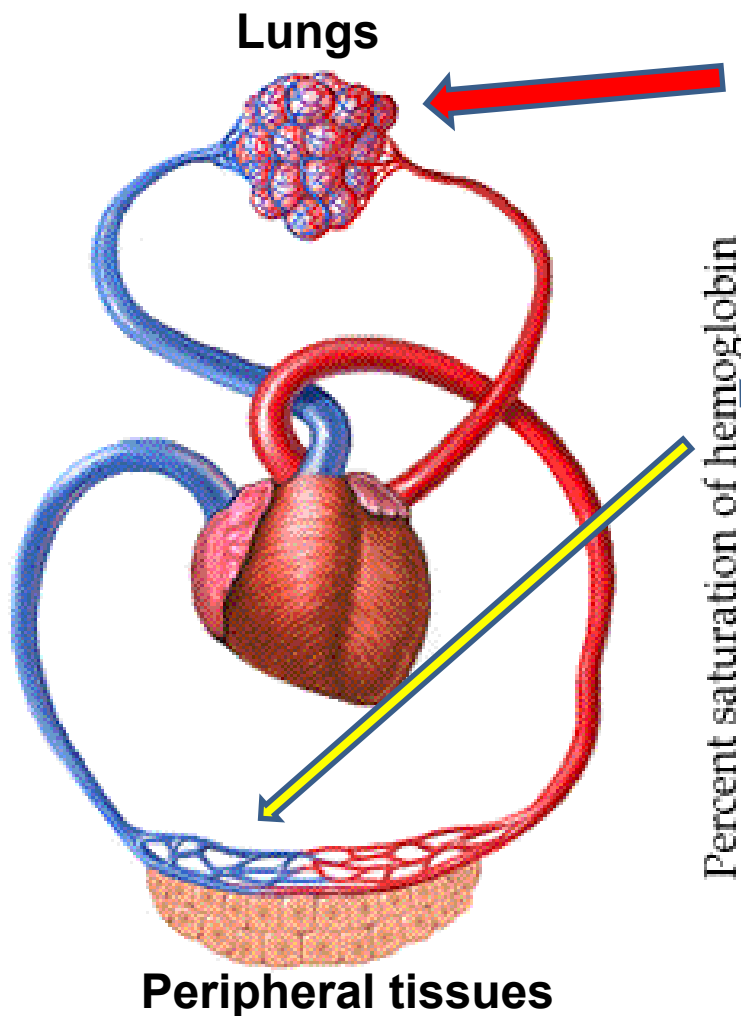
Oxygen binds rapidly and reversibly to hemoglobin:



The amount of oxyhemoglobin (percentage saturation of hemoglobin with oxygen) is a function of the partial pressure of oxygen in the blood

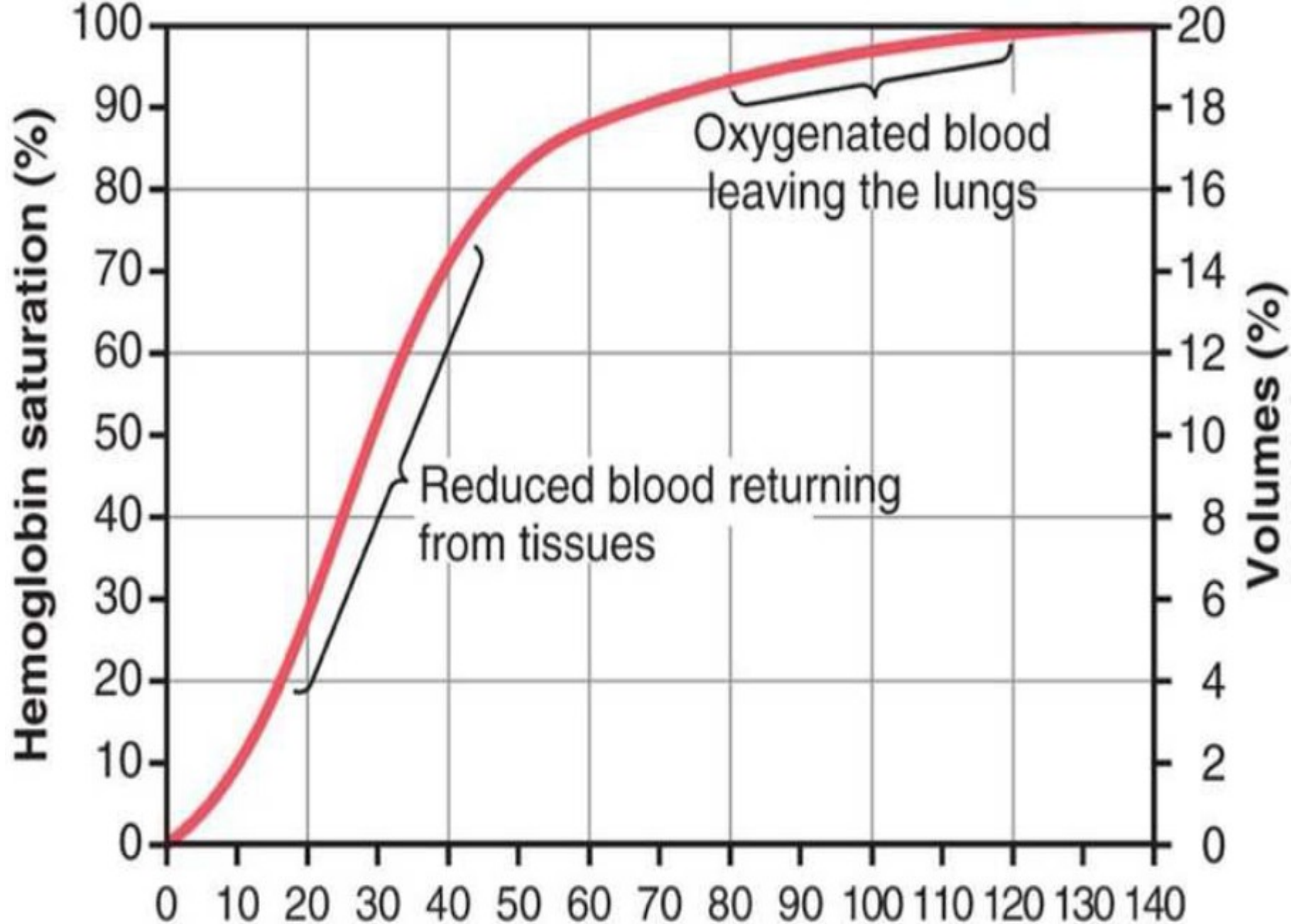
Oxygen-hemoglobin dissociation (association) curve is a curve which demonstrates a progressive increase in the percentage of hemoglobin bound with oxygen as blood PO_2 increases, *and vice versa*, which is called the *percent saturation of hemoglobin*

OXYGEN-HEMOGLOBIN DISSOCIATION CURVE



The blood leaving the lungs and entering the systemic arteries usually has a PO_2 of about 95 mm Hg thus the *usual oxygen saturation of systemic arterial blood averages 97 percent*

In **venous blood** returning from the peripheral tissues, the PO_2 is about 40 mm Hg, and *the saturation of hemoglobin averages 75 percent*



O₂ dissociation/association curve is S-shaped curve

In the S-shaped curve of the O₂ dissociation curve:

The plateau region of the curve is the loading phase; this flat upper portion means that even if the P_{O₂} in alveolar gas falls (changes) some what, loading of O₂ will be little affected

The steep lower part (unloading phase) of the dissociation curve means that the peripheral tissues can withdraw large amounts of O₂ for only a small drop in capillary P_{O₂}

Factors That Shift the Oxygen-Hemoglobin Dissociation/association Curve

Shifting the O_2 dissociation curve to the right or to the left of normal means changing in the binding affinity of hemoglobin for O_2

O_2 dissociation curve shifted to the right indicates the O_2 affinity of Hb is reduced

O_2 dissociation curve shifted to the left signifies an increase in hemoglobin's affinity for oxygen

Factors that shift the O₂ dissociation curve to the right or to the left

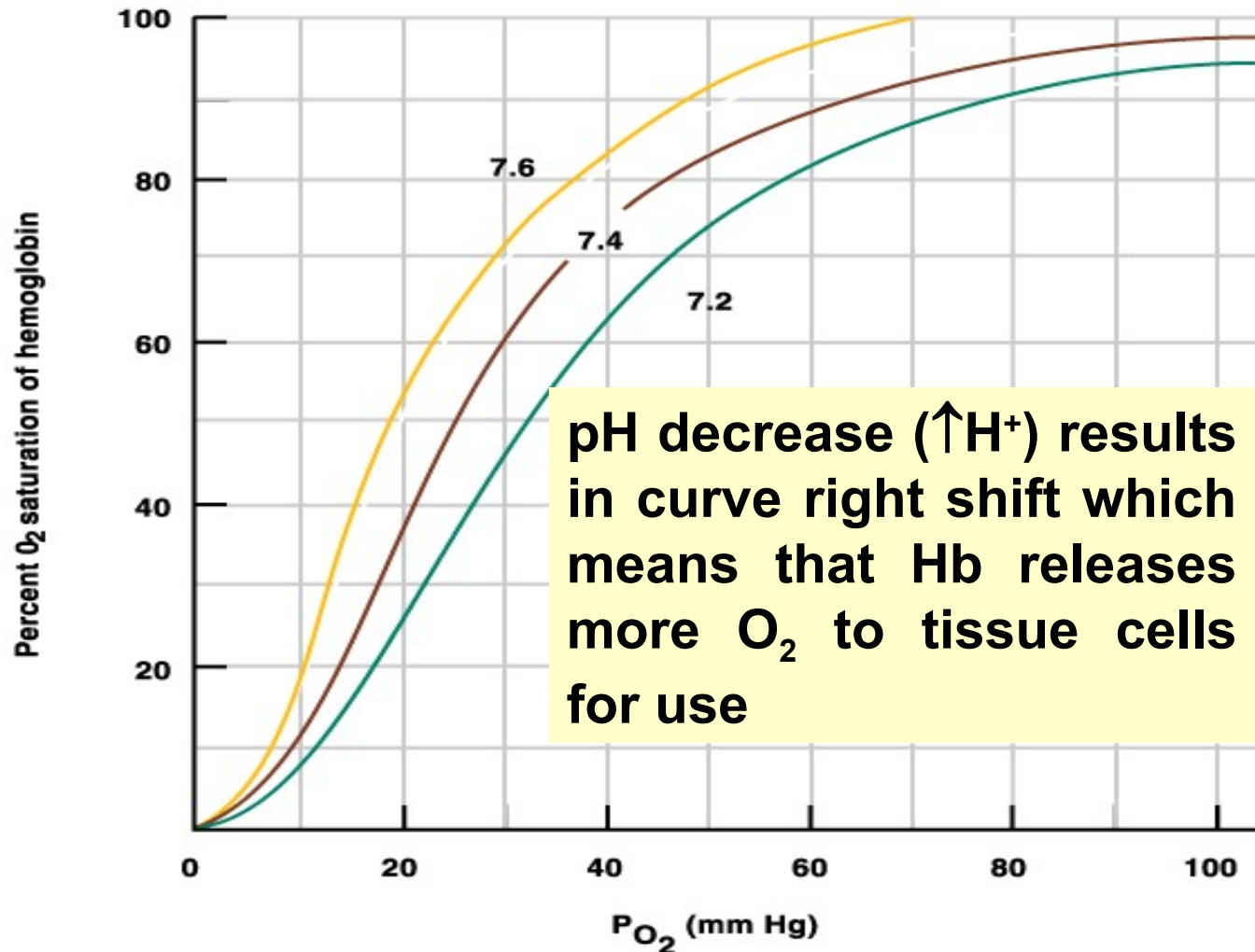
- An increase/decrease in H⁺ concentration
- An increase/decrease P_{CO2}
- An increase/decrease in temperature, and
- An increase/decrease the concentration of 2,3-diphosphoglycerate (DPG)

(also known as 2,3-biphosphoglycerate;BPG),

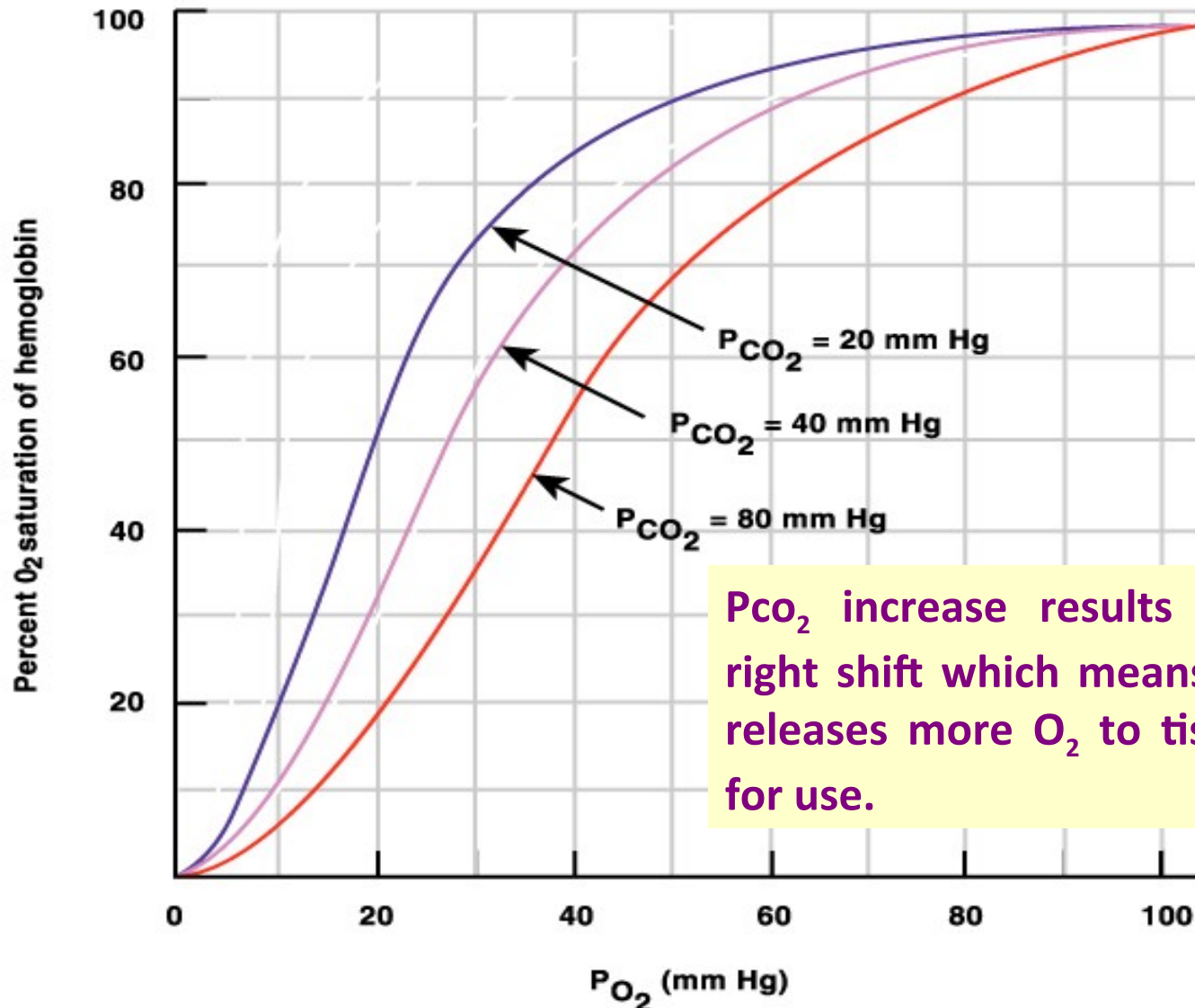
- * increase of each of the above factor in the red cells shift the O₂ dissociation curve to the right
- * opposite changes of the factors in RBCs shift the curve to the left

Effect of pH (H^+ concentration) on oxygen dissociation curve

Effect of pH



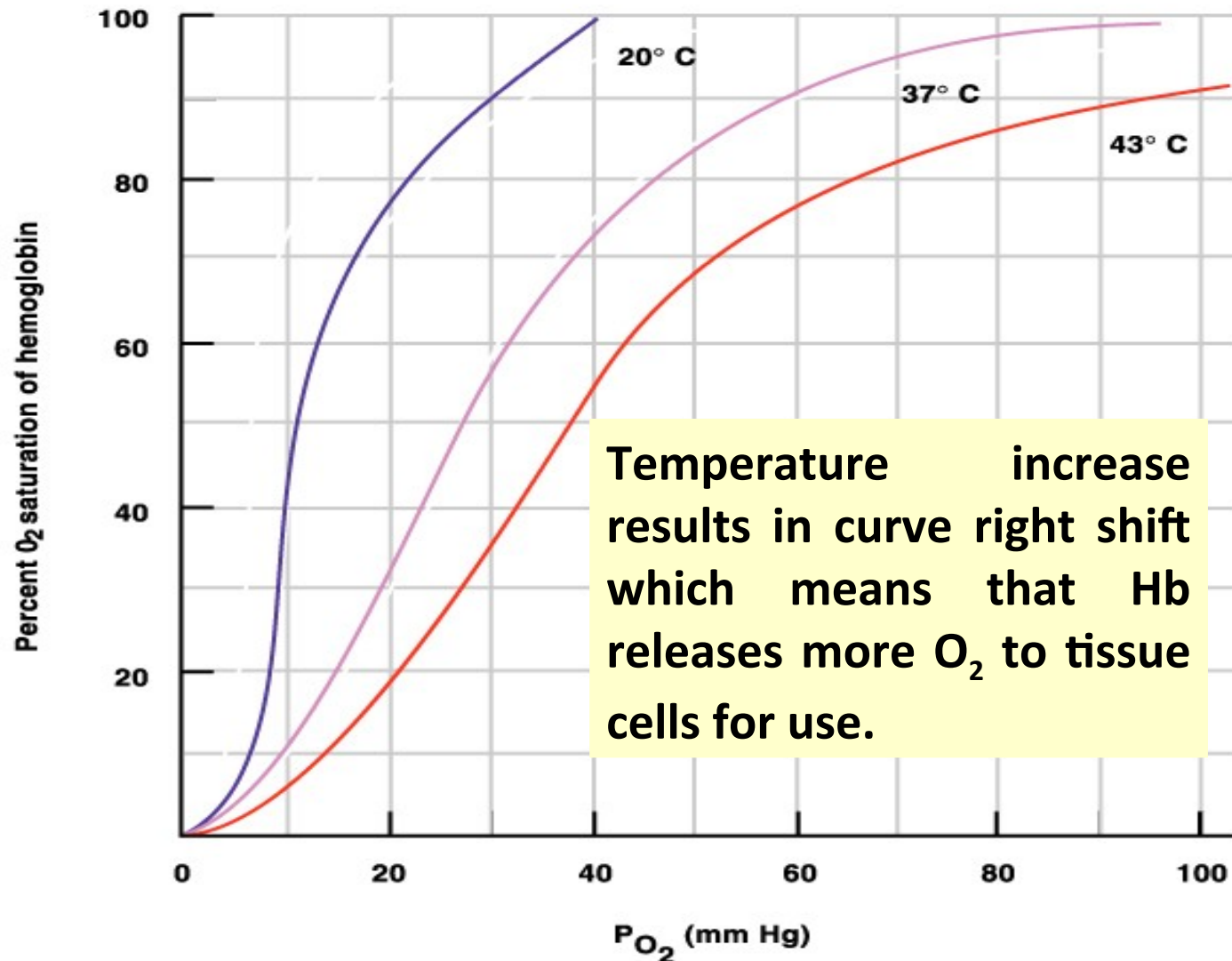
Effect of P_{CO_2} on oxygen dissociation curve



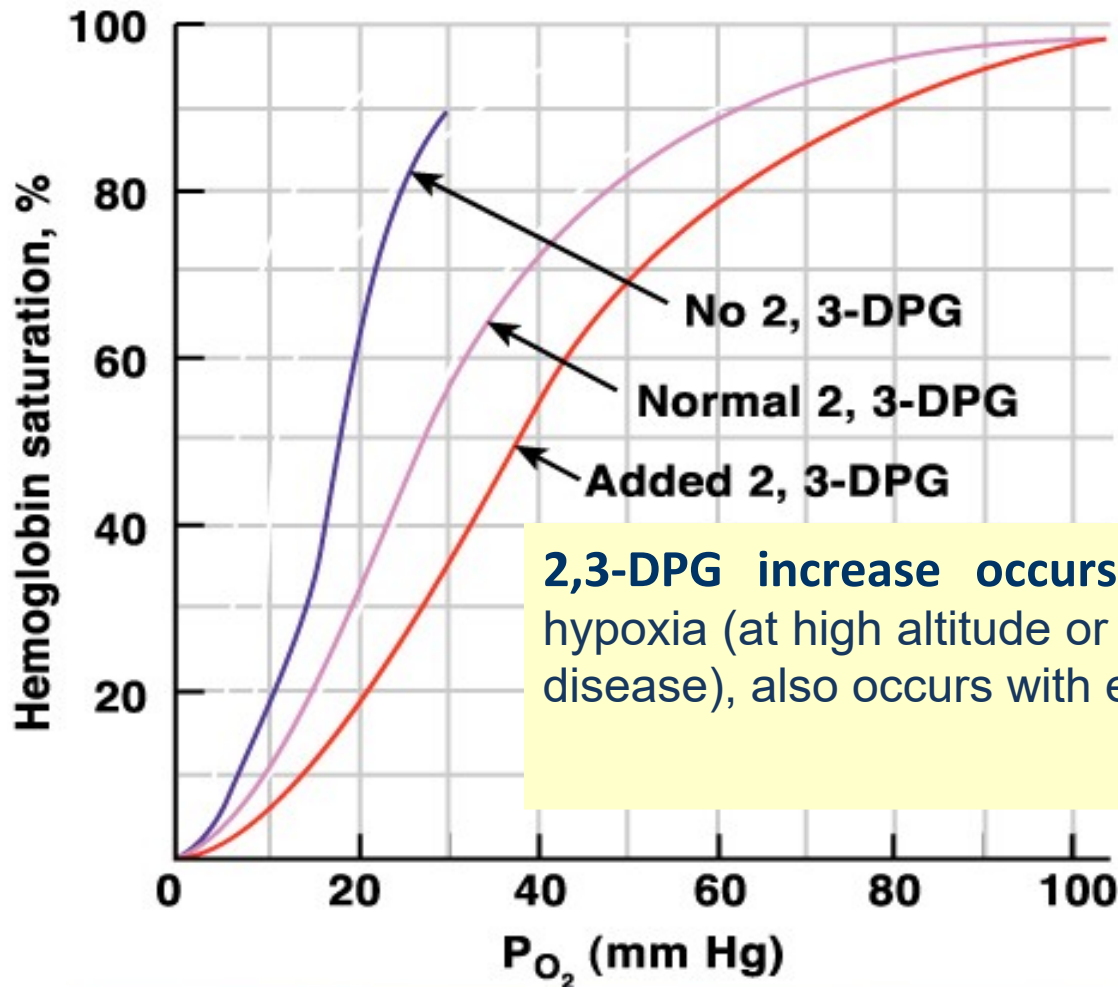
P_{CO_2} increase results in curve right shift which means that Hb releases more O_2 to tissue cells for use.

Effect of temperature on oxygen dissociation curve

Effect of temperature



Effect of 2,3-DPG on oxygen dissociation curve



Graph question: Blood stored in blood banks loses its normal content of 2, 3-DPG. Is this

The Bohr Effect

A shift of the oxygen-hemoglobin dissociation curve to the right in response to *increases in blood carbon dioxide and hydrogen ions* enhances the release of oxygen from the blood in the tissues

This is called the ***Bohr effect***

In peripheral tissues blood PCO_2 rises, which in turn raises the blood H_2CO_3 and the hydrogen ion concentration; these shift the oxygen-hemoglobin dissociation curve to the right and downward; forcing oxygen away from the hemoglobin and therefore delivering increased amounts of oxygen to the tissues

Carbon Monoxide (CO) Competes With Oxygen's Binding of Hemoglobin

- Carbon monoxide binds to hemoglobin to form carboxyhemoglobin (HbCO)
$$\text{Hb} + \text{CO} \leftrightarrow \text{HbCO}$$
- Carbon monoxide interferes with oxygen transport by competing for the same binding sites on hemoglobin
- Hemoglobin affinity to CO is about 210 times that of its affinity to oxygen
- So breathing a small amount of CO can result in the formation of large amounts of HbCO

Presence of CO greatly reduces oxygen content of blood because oxygen cannot bind to hemoglobin once it forms bound CO

Presence of COHb also shifts the O₂ dissociation curve to the left, thus interfering with the unloading of O₂

Carbon monoxide is dangerous for several reasons:

- It has a strong binding affinity for hemoglobin
- As an odorless, colorless, and nonirritating gas, it is virtually undetectable

- PaO₂ is normal in person exposed to high concentration of CO, and there is no feedback mechanism to indicate that oxygen content is low
- There are no physical signs of hypoxemia (i.e., cyanosis or bluish color around the lips and fingers)

because the blood is bright cherry red when CO binds with hemoglobin

The treatment for CO poisoning is breathing 100% oxygen

The addition of 5% carbon dioxide to the inspired gas stimulates ventilation, which lowers the CO and

Transport of Carbon Dioxide in the Blood

On average 4 ml of carbon dioxide is transported from the tissues to the lungs in each 100 ml of blood

Carbon dioxide is carried in the blood in three forms:

- I) Dissolved in the plasma (7%)
- II) As bicarbonate ions in the plasma and in the red cells (70%)
- III) As carbamino proteins (mainly carbaminohemoglobin) (23%)

i) Dissolved in the plasma (7%)

The amount of carbon dioxide dissolved in the fluid of the blood at 45 mm Hg is about 2.7 ml/dl (2.7 volumes percent) (systemic capillaries)

The amount of carbon dioxide dissolved at 40 mm Hg is about 2.4 ml (pulmonary capillaries)

Thus, about 0.3 ml (2.7ml -2.4ml) of carbon dioxide is transported in the dissolved form by each 100 ml of blood flow from peripheral tissues to the lungs

ii) As bicarbonate ions in the plasma and in the red cells (70%)

Carbon dioxide diffuses from tissue cells in to erythrocytes of systemic capillaries blood

In the RBCs the following reaction occurs:



CA=carbonic anhydrase

CA accelerates the reaction rate about 5000-fold

Most of the H^+ ions liberated are bound to Hb

As H^+ binds to hemoglobin, it decreases oxygen binding affinity of hemoglobin and shifts the oxyhemoglobin equilibrium curve to the right

This promotes the unloading of oxygen from hemoglobin in the tissues

Many of the HCO_3^- ions diffuse from the red cells into the plasma

Chloride ions diffuse into the red cells to replace HCO_3^- ions

Bicarbonate-chloride carrier protein in the red cell membrane shuttles these two ions in opposite directions at rapid velocities

This exchange phenomenon is called *the chloride shift*

In pulmonary capillaries the above processes are reversed and carbon dioxide released from the blood in to alveoli

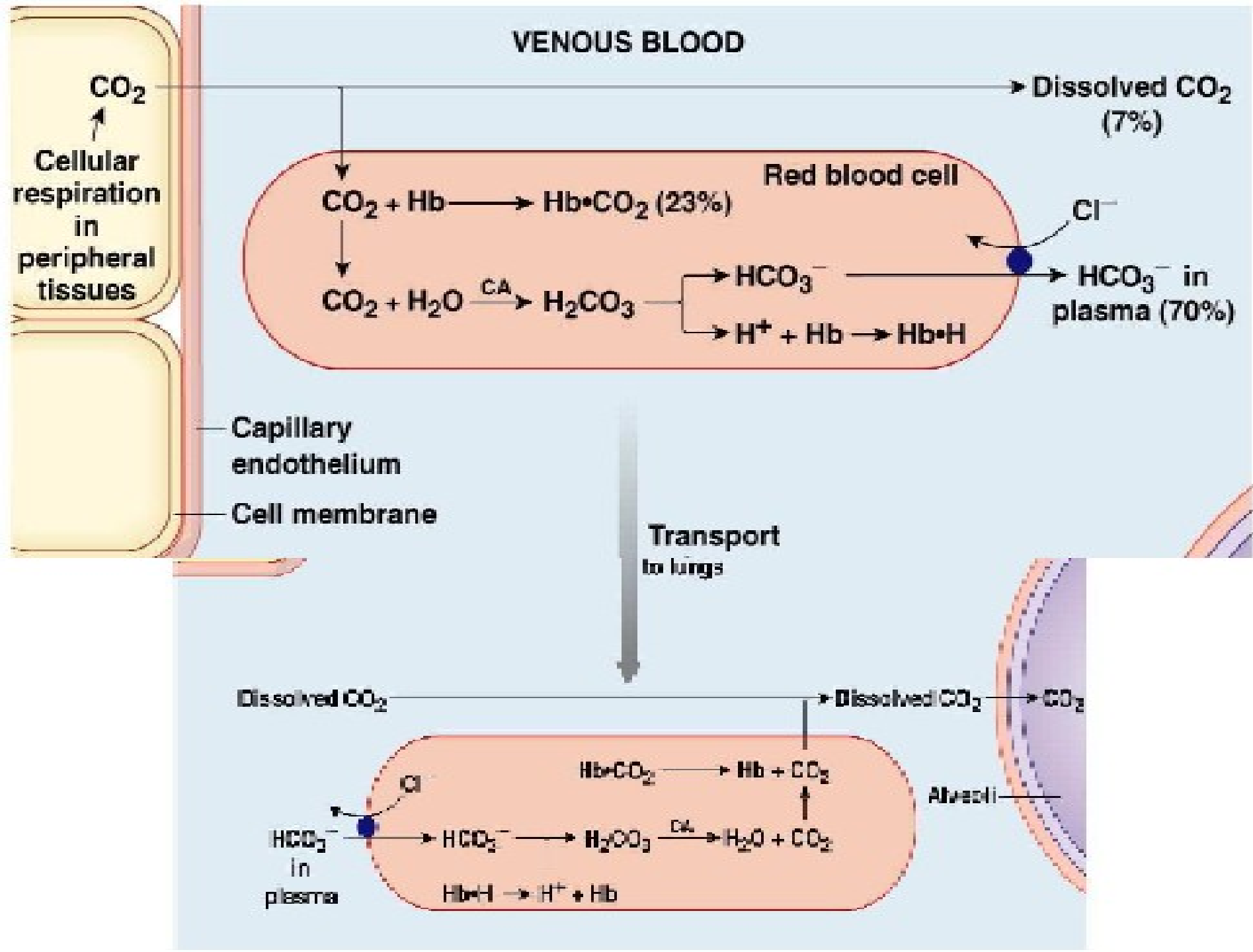
The chloride content of venous red blood cells is greater than that of arterial red cells (because of chloride shift)

The events associated with the uptake of CO_2 by blood increase the osmolar content of the red cell, and, consequently, water enters the cell, thus increasing its volume

When the cells pass through the lung, they shrink a little

This mechanism of transport of CO_2 (in the form of HCO_3^- ions) accounts for about **70% of the CO_2** transported from the tissues to the lungs

When experimentally a carbonic anhydrase inhibitor (acetazolamide) is administered to an animal to block the action of carbonic anhydrase in the red blood cells, carbon dioxide transport from the tissues becomes so poor that the tissue PCO_2 can be made to **rise to 80 mm Hg** instead of the normal 45 mm Hg



iii) As carbamino proteins (mainly carbaminohemoglobin) (23%)

CO₂ reacts directly with the hemoglobin molecule to form the compound *carbaminohemoglobin* (CO₂Hb)

This reaction occurs rapidly **without an enzyme**, and reduced Hb can bind more CO₂ as carbamino-hemoglobin than HbO₂

Thus, unloading of O₂ in peripheral capillaries facilitates the loading of CO₂, whereas oxygenation has the opposite effect

A small amount of CO₂ also reacts in the same way with the plasma proteins in the tissue capillaries

This is much less significant for the transport of carbon dioxide because the quantity of these proteins in the blood is only one fourth as great as the quantity of hemoglobin

The Haldane Effect

The Haldane effect states that deoxygenated Hb has a greater affinity for CO_2 than does O_2

Thus, O_2 release at the tissues facilitates CO_2 pickup while O_2 pickup at the lungs facilitates CO_2 release

The *Haldane effect*, is quantitatively far more important in promoting carbon dioxide transport than is the Bohr effect in promoting oxygen transport

5) Regulation of Respiration

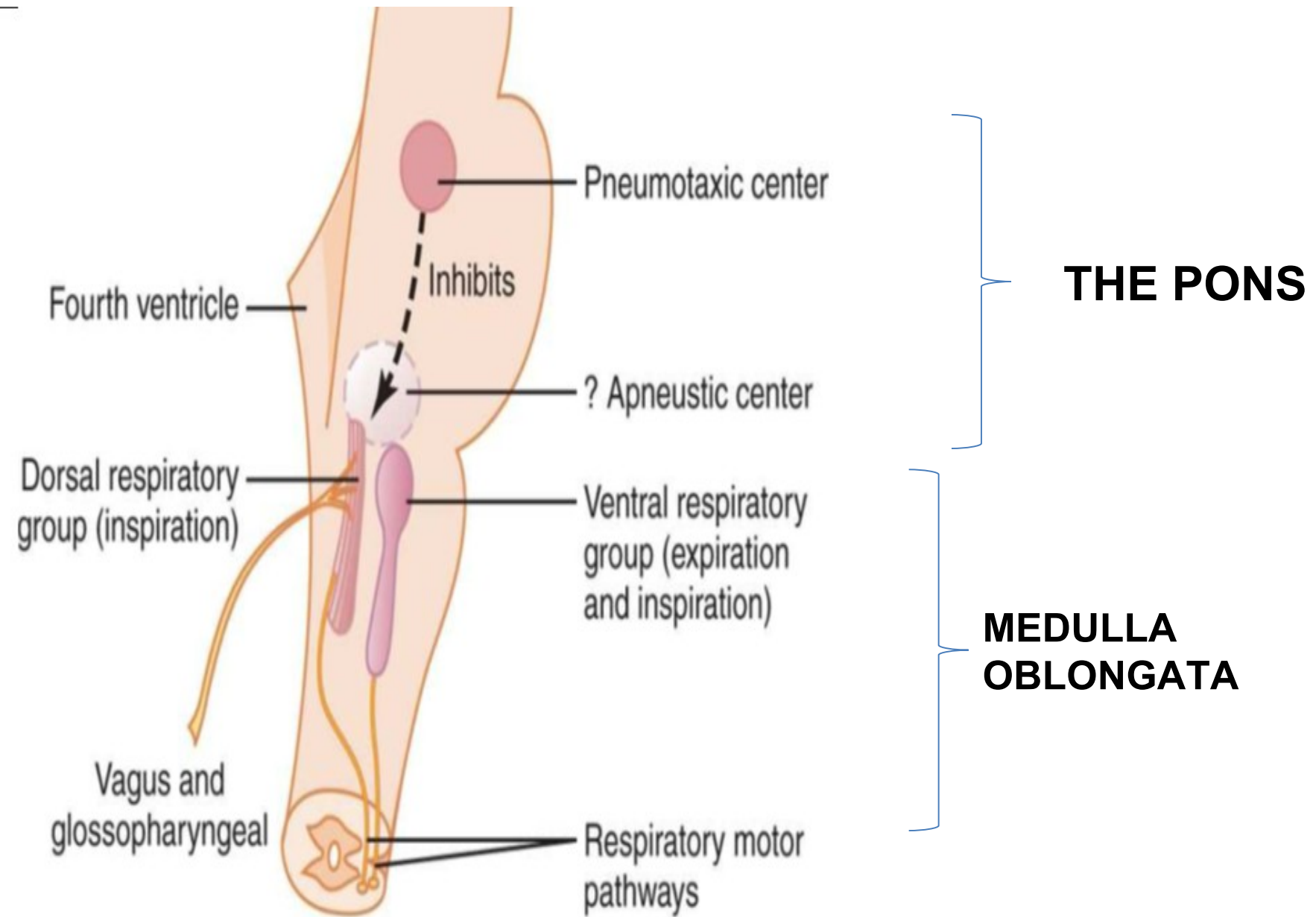
The nervous system adjusts the rate of alveolar ventilation almost exactly to the demands of the body

So, that the oxygen pressure (PO_2) and carbon dioxide pressure (PCO_2) in the arterial blood are hardly altered even during heavy exercise

Respiratory Center

The *respiratory center*:

- Is composed of several groups of neurons located *bilaterally* in the *medulla oblongata* and *pons* of the brain stem
- Has *three major collections of neurons*:
 - i) *Dorsal respiratory group of neurons (DRG)*, located in the dorsal portion of the medulla
 - ii) *Ventral respiratory group of neurons (VRG)*, located in the ventrolateral part of the medulla
 - iii) *The pneumotaxic center*, located dorsally in the superior portion of the pons



DORSAL RESPIRATORY GROUP OF NEURONS

Most of its neurons are located within the *nucleus of the tractus solitarius (NTS)*, although additional neurons located in the adjacent reticular substance of the medulla

The NTS is the sensory termination of both the vagal and the glossopharyngeal nerves, which transmit sensory signals into the respiratory center from

- (1) Peripheral chemoreceptors,
- (2) Baroreceptors, and
- (3) Several types of receptors in the lungs

Dorsal Respiratory Group of Neurons (I-neurons)

It controls *inspiration* and *respiratory rhythm*

The basic rhythm of respiration is generated in the dorsal respiratory group of neurons

It emits repetitive nerve signals that are *inspiratory neuronal action potentials* that causes rhythmical inspiration

Nervous signal that begins weakly and increases steadily in a ramp manner for about 2 seconds is transmitted to the inspiratory muscles (mainly diaphragm) to cause inspirations from this group of neurons

Then, it ceases abruptly for approximately the next 3 seconds, which turns off the excitation of the diaphragm and allows elastic recoil of the lungs and the chest wall to cause expiration

Next, the inspiratory signal begins again for another cycle; this cycle repeats again and again, with expiration occurring in between

Pneumotaxic Center

- Located dorsally in the *nucleus parabrachialis* of the upper pons
- Transmits signals to the inspiratory area (DRG)
- Primary effect of this center is to control the "switch-off" point of the inspiratory ramp (inspiratory signal of the dorsal respiratory group of neurons), thus controlling the duration of the filling phase of the lung cycle

This limiting inspiration effect of pneumotaxic center has a secondary effect of increasing the rate of breathing because limitation of inspiration also shortens expiration and the entire period of each respiration

Strong pneumotaxic signal to DRG can increase the rate of breathing to 30 to 40 breaths per minute, whereas a weak pneumotaxic signal may reduce the rate to only 3 to 5 breaths per minute

Ventral Respiratory Group of Neurons (E-neurons)

- Has both inspiration and expiration function
- Located in each side of the medulla anteriolateral to the dorsal respiratory group of neurons
- Found in the *nucleus ambiguus* rostrally and the *nucleus retroambiguus* caudally
- It remains almost totally *inactive* during normal quiet respiration

The ventral respiratory neurons do not appear to participate in the basic rhythmical oscillation that controls respiration

When the respiratory drive for increased pulmonary ventilation becomes greater than normal, the ventral respiratory area becomes active and causes extra respiratory drive

Electrical stimulation of a few of the neurons in the ventral group causes inspiration, whereas stimulation of others causes expiration

Therefore, these neurons contribute to both inspiration and expiration

Chemical Control of Respiration

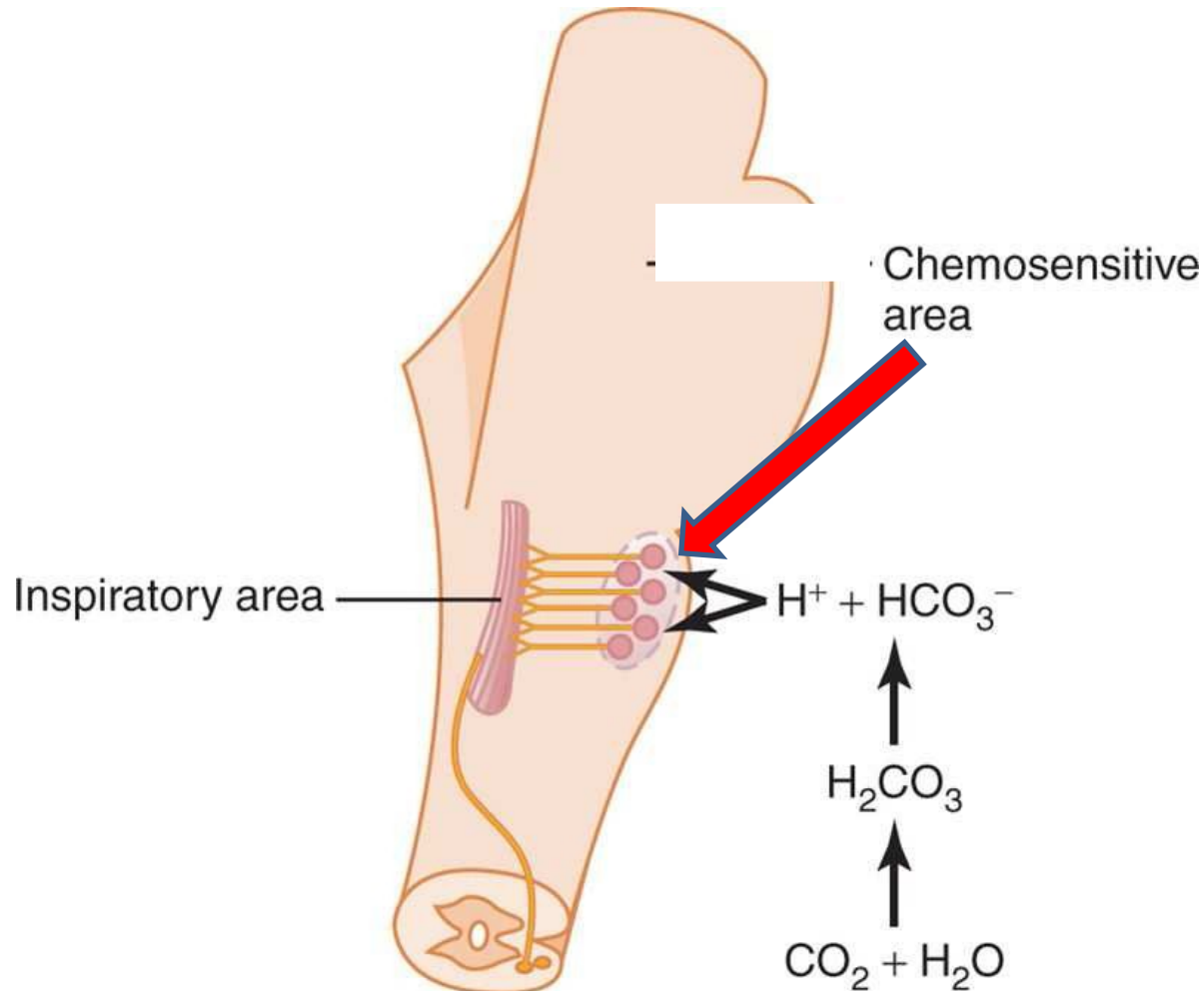
The respiratory function is increased or decreased to match the ventilatory needs of the body (it is regulated process)

i) Central chemoreceptors

Chemical control of respiratory center activity by CO_2 and H^+

- There is *a chemosensitive neuronal area* lying in the ventral surface of the medulla (central chemoreceptors)
- This area is highly sensitive to changes in either

Excitation of the chemosensitive neurons in turn excites the other portions of the respiratory center that in turn increases respiration



Hall: Guyton and Hall Textbook of Medical Physiology, 12th Edition
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Excitation of the chemosensitive neurons by hydrogen ions is likely the primary stimulus

However, hydrogen ions do not easily cross the blood-brain barrier

For this reason, change in hydrogen ion concentration in the blood has considerably less effect in stimulating the chemosensitive neurons than does change in blood carbon dioxide, even though carbon dioxide is believed to stimulate these neurons secondarily by changing the hydrogen ion concentration

Whenever the blood PCO_2 increases, so does the PCO_2 of both the interstitial fluid of the medulla and the cerebrospinal fluid because carbon dioxide passes through blood-brain barrier almost as if the barrier does not exist

In both these fluids, the carbon dioxide immediately reacts with the water to form new hydrogen ions

Thus, more hydrogen ions are released into the respiratory chemosensitive sensory area of the medulla and this increases respiratory center activity

Excitation of the respiratory center by carbon dioxide is great the first few hours after the blood carbon dioxide first increases, but then it gradually declines over the next 1 to 2 days, decreasing to about one-fifth the initial effect

Part of this decline results from renal readjustment of the hydrogen ion concentration in the circulating blood back toward normal after the carbon dioxide first increases the hydrogen concentration

The kidneys achieve this by increasing the blood bicarbonate, which binds with the hydrogen ions in the blood and cerebrospinal fluid to reduce their concentrations

Moreover, over a period of hours, the bicarbonate ions also slowly diffuse through the blood-brain and blood-cerebrospinal fluid barriers and combine directly with the hydrogen ions adjacent to the respiratory neurons as well, thus reducing the hydrogen ions back to near normal. The kidneys also secrete excess H^+ ions to eliminate them with the urine eventually.

A change in blood carbon dioxide concentration therefore has a potent *acute* effect on controlling respiratory drive but only a weak *chronic* effect after a few days adaptation

ii) Peripheral Chemoreceptor

Role of Oxygen in Respiratory Control

Special nervous chemical receptors, called peripheral *chemoreceptors*, are located in several areas outside the brain

Most of the chemoreceptors are in the *carotid bodies*

The *carotid bodies* are located bilaterally in the bifurcations of the common carotid arteries

Their afferent nerve fibers pass through Hering's nerves to the *glossopharyngeal nerves* and then to the dorsal respiratory group area of the medulla

However, a few peripheral chemoreceptors are also in the *aortic bodies* and a very few are located elsewhere in association with other arteries of the thoracic and abdominal regions

The *aortic bodies* are located along the arch of the aorta; their afferent nerve fibers pass through the *vagi*, also to the medullary dorsal respiratory group area

Peripheral chemoreceptors are especially important for detecting changes in oxygen in the blood, although they also respond to a lesser extent to changes in CO₂ and H⁺

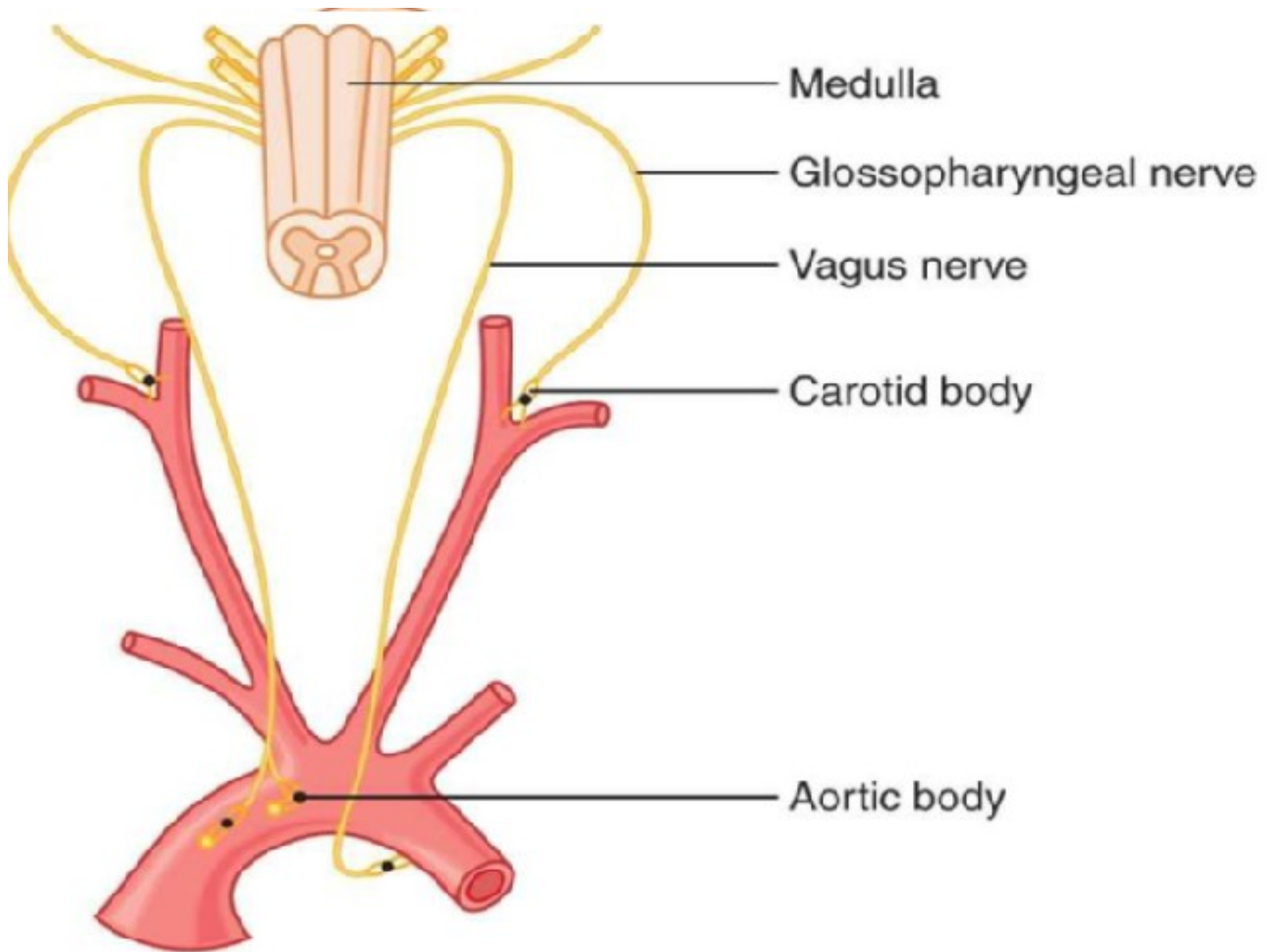


FIG. PERIPHERAL CHEMORECEPTORS

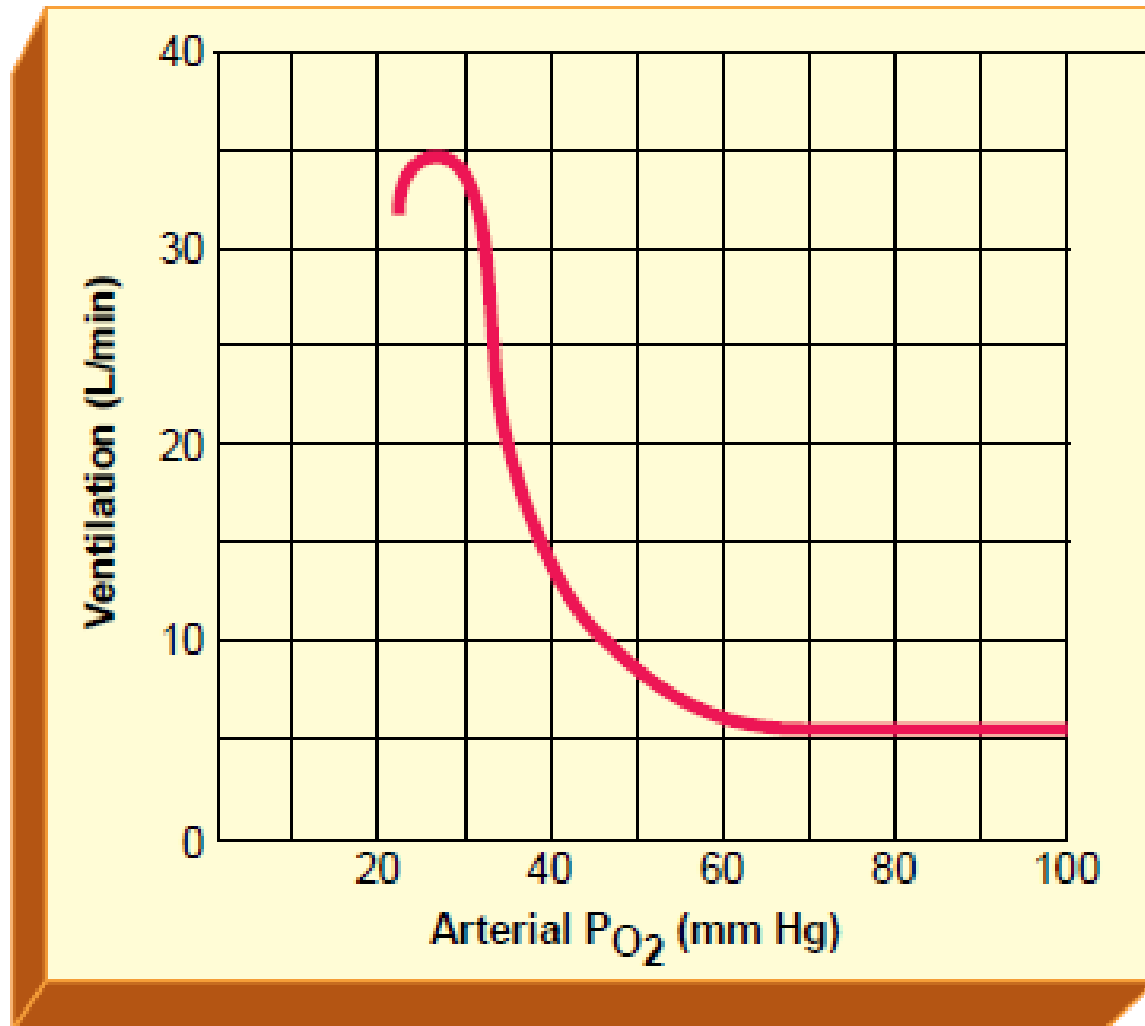
Blood flow through these *carotid* and *aortic* bodies is very high, 20 times the weight of the bodies themselves each minute

Therefore, the percentage of oxygen removed from the flowing blood is virtually zero in these carotid and aortic bodies

This means that *the chemoreceptors are exposed at all times to arterial blood*, not venous blood, and their PO_2 are arterial PO_2

When the oxygen concentration in the arterial blood falls below normal, the chemoreceptors become strongly stimulated

The effect of low levels of arterial P_{O_2} on ventilation



↓ Arterial P_{O_2}



Peripheral Chemoreceptors
↑ nerve impulses



Respiratory Centers
↑ nerve impulses



Respiratory Muscles
↑ contractions

↑ Ventilation

Chronic Breathing of Low Oxygen Stimulates Respiration Even More (The Phenomenon of "Acclimatization")

Mountain climbers have found that when they ascend a mountain slowly, over a period of days rather than a period of hours, they breathe much more deeply and therefore can withstand far lower atmospheric oxygen concentrations than when they ascend rapidly

This is because of *acclimatization*

The reason for acclimatization is that, within 2 to 3 days, the respiratory center in the brain stem loses about four fifths of its sensitivity to changes in PCO_2 and hydrogen ions

Therefore, the excess ventilatory blow-off of carbon dioxide that normally would inhibit an increase in respiration fails to occur, and low oxygen can drive the respiratory system to a much higher level of alveolar ventilation than under acute conditions

Increased Carbon Dioxide and Hydrogen Ion Concentration Stimulates the Peripheral Chemoreceptors

An increase in either carbon dioxide concentration or hydrogen ion concentration also excites the peripheral chemoreceptors and increases respiratory activity

However, the effects of both these factors in the central chemoreceptors are much more powerful than their effects mediated through the peripheral chemoreceptors (about seven times as powerful)

The other difference between the peripheral and central effects of carbon dioxide is that the stimulation by way of the peripheral chemoreceptors occurs as much as five times as rapidly as central stimulation

6) Respiratory insufficiency

Respiratory insufficiency is the condition in which the lungs cannot take in sufficient oxygen and/or expel sufficient carbon dioxide to meet the needs of the cells of the body

Hypoxia is a condition in which the body as a whole (*generalized hypoxia*) or a region of the body (*tissue hypoxia*) is deprived of adequate oxygen supply

Hypoxemia /hypoxaemia/ is a decreased partial pressure of oxygen in arterial blood; to less than 60 mmHg or causing hemoglobin oxygen saturation of less than 90%

Causes of hypoxia:

1) Inadequate oxygenation of the blood in the lungs because of extrinsic reasons

- Deficiency of oxygen in the atmosphere
 - Hypoventilation (neuromuscular disorders)

2) Pulmonary disease

- Hypoventilation caused by increased airway resistance or decreased pulmonary compliance
- Abnormal alveolar ventilation-perfusion ratio (including either increased physiologic dead space or increased physiologic shunt)
- Diminished respiratory membrane diffusion

- 3) Venous-to-arterial shunts ("right-to-left" cardiac shunts)
- 4) Inadequate oxygen transport to the tissues by the blood

- Anemia or abnormal hemoglobin
- General circulatory deficiency
 - Localized circulatory deficiency (peripheral, cerebral, coronary vessels)
- Tissue edema

5) Inadequate tissue capability of using oxygen

- Poisoning of cellular oxidation enzymes
 - Diminished cellular metabolic capacity for using oxygen, because of toxicity, vitamin deficiency, or other factors

Oxygen Therapy in Different Types of Hypoxia

Oxygen can be administered by

- 1) placing the patient's head in a "tent" that contains air fortified with oxygen,
- 2) allowing the patient to breathe either pure oxygen or high concentrations of oxygen from a mask, or
- 3) administering oxygen through an intranasal tube

In *atmospheric hypoxia*, oxygen therapy can completely correct the depressed oxygen level in the inspired gases and, therefore, provide 100 percent effective therapy

In *hypoventilation hypoxia*, a person breathing 100 percent oxygen can move five times as much oxygen into the alveoli with each breath as when breathing normal air

Therefore, here again oxygen therapy can be extremely beneficial

However, this provides no benefit for the excess blood carbon dioxide also caused by the hypoventilation

In hypoxia caused by impaired alveolar membrane diffusion, essentially the same result occurs as in hypoventilation hypoxia

In hypoxia caused by anemia, abnormal hemoglobin transport of oxygen, circulatory deficiency, or physiologic shunt, oxygen therapy is of much less value because normal oxygen is already available in the alveoli

The problem, in these above situations, instead is that one or more of the mechanisms for transporting oxygen from the lungs to the tissues are deficient

Even so, a small amount of extra oxygen, between 7 and 30 percent, can be *transported in the dissolved state* in the blood when alveolar oxygen is increased to maximum even though the amount transported by the hemoglobin is hardly altered

This small amount of extra oxygen may be the difference between life and death

In the different types of *hypoxia caused by inadequate tissue use of oxygen*, there is abnormality neither of oxygen pickup by the lungs nor of transport to the tissues

Instead, the tissue metabolic enzyme system is simply incapable of using the oxygen that is delivered

Therefore, oxygen therapy provides no measurable benefit in these cases

Cyanosis

Cyanosis is blueness of the skin due to excessive amounts of deoxygenated hemoglobin in the skin blood vessels, especially in the capillaries

Deoxygenated hemoglobin has an intense dark blue-purple color that is transmitted through the skin

Definite cyanosis appears whenever the *arterial blood* contains more than 5 grams of deoxygenated hemoglobin in each 100 milliliters of blood

Hypercapnia

Hypercapnia means excess carbon dioxide in the body fluids

Hypercapnia occurs in association with hypoxia when the hypoxia is caused by hypoventilation or circulatory deficiency

When the alveolar P_{CO_2} rises above about 60 to 75 mm Hg, an otherwise normal person by then is breathing about as rapidly and deeply as he or she can, and “air hunger,” also called dyspnea, becomes severe

If the P_{CO_2} rises to 80 to 100 mm Hg, the person becomes lethargic and sometimes even semicomatose

Anesthesia and death can result when the P_{CO_2} rises to 120 to 150 mm Hg

Dyspnea

Dyspnea means mental anguish associated with inability to ventilate enough to satisfy the demand for air

A common synonym is air hunger

At least three factors often cause the sensation of dyspnea:

- i) abnormality of respiratory gases in the body fluids, hypercapnia and, to a much less extent, hypoxia
- ii) the amount of work that must be performed by the respiratory muscles to provide adequate ventilation
- iii) state of mind

Restrictive and Obstructive Lung Defects

Restrictive disorders are those in which expansion of the lung is restricted, either due to loss of alveolar volume, or diseases of the chest wall, pleura or neuromuscular apparatus

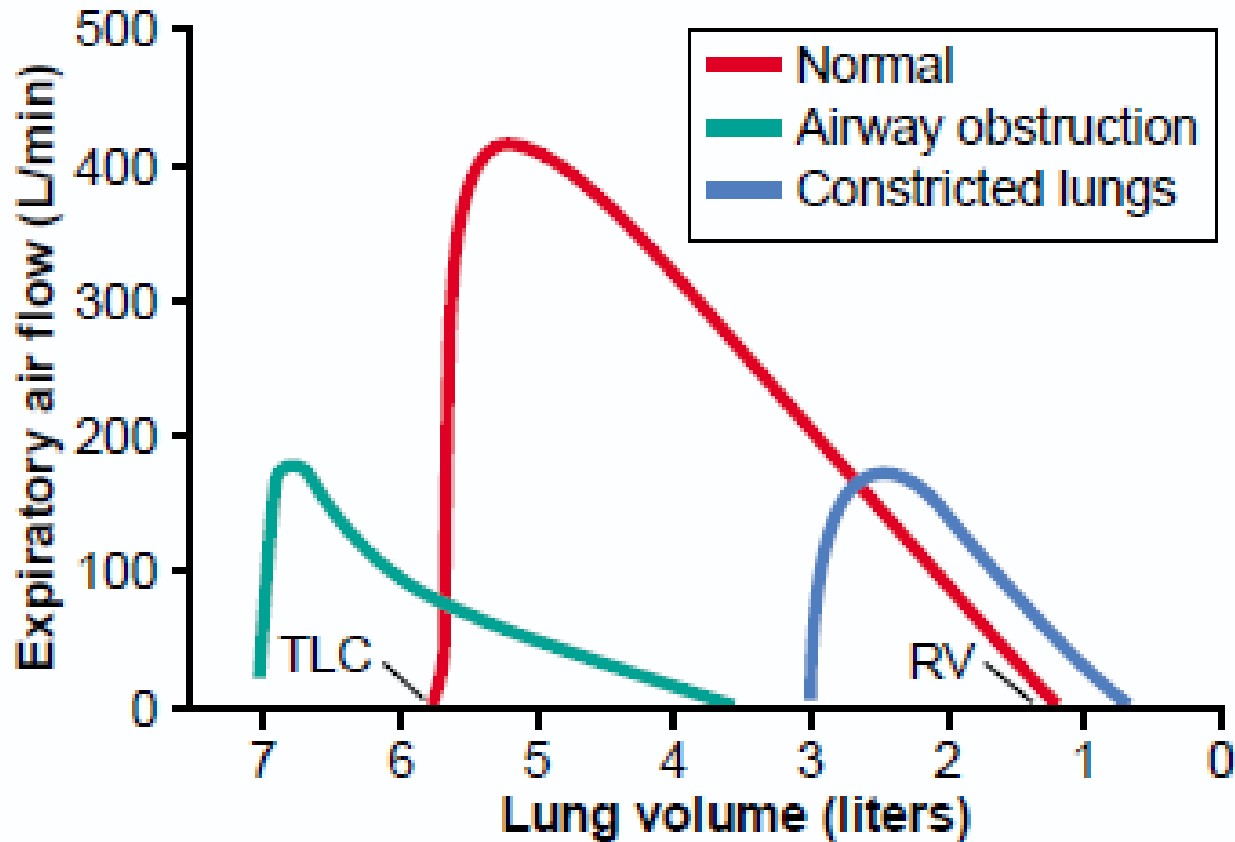
Constricted lung diseases include fibrotic diseases of the lung itself, such as *tuberculosis* and *silicosis*, and diseases that constrict the chest cage, such as *kyphosis*, *scoliosis*, and *fibrotic pleurisy*

Obstructive diseases occur when airway disease causing airflow limitation that slows lung emptying

In diseases with *airway obstruction*, it is usually much more difficult to expire than to inspire because the closing tendency of the airways is greatly increased by the extra positive pressure required in the chest to cause expiration

The classic disease that causes severe airway obstruction is *asthma*

Serious airway obstruction also occurs in some stages of *emphysema*



Effect of two respiratory abnormalities; constricted lungs and airway obstruction; on the maximum expiratory flow-volume curve, total lung capacity (TLC), and residual volume (RV).

Physiologic Peculiarities of Specific Pulmonary Abnormalities

i) Chronic Pulmonary Emphysema

Pulmonary emphysema literally means excess air in the lungs

Pulmonary emphysema describes complex obstructive and destructive process of the lungs caused by many years of smoking

It results from the following major pathophysiologic changes in the lungs:

1. *Chronic infection*, caused by inhaling smoke or other substances that irritate the bronchi and bronchioles

The chronic infection seriously deranges the normal protective mechanisms of the airways, including partial paralysis of the cilia of the respiratory epithelium, an effect caused by nicotine

As a result, mucus cannot be moved easily out of the passageways

Also, stimulation of excess mucus secretion occurs, which further exacerbates the condition

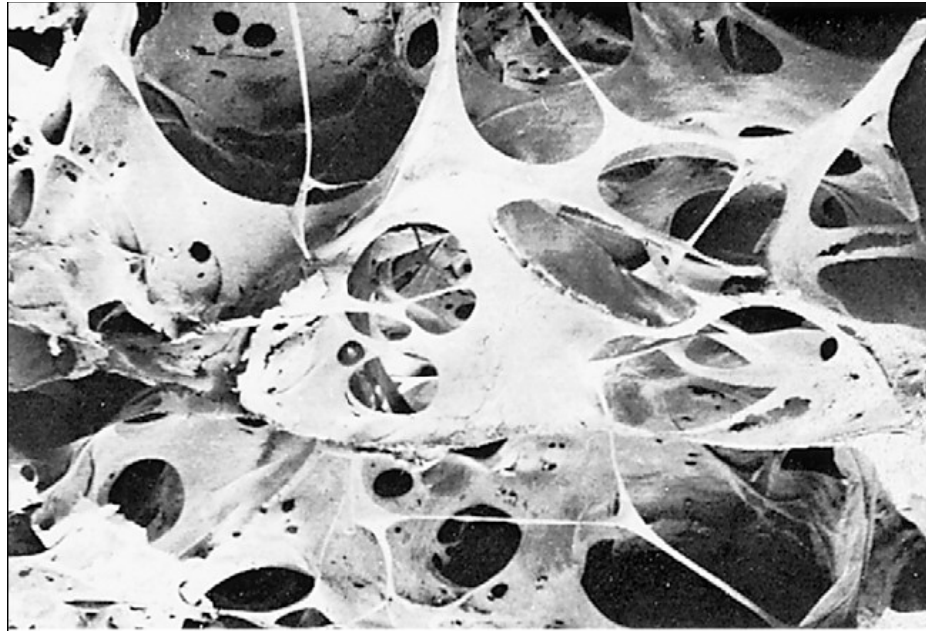
Too, inhibition of the alveolar macrophages occurs, so that they become less effective in combating infection

2. The infection, excess mucus, and inflammatory edema of the bronchiolar epithelium together cause *chronic obstruction* of many of the smaller airways

3. The obstruction of the airways makes it especially difficult to expire, thus causing *entrapment of air in the alveoli* and overstretching them

This, combined with the lung infection, causes *marked destruction of as much as 50 to 80 % of the alveolar walls*

Figure
Contrast of the
emphysematous lung
(*top figure*) with the
normal lung (*bottom
figure*), showing
extensive alveolar
destruction in
emphysema.



The physiologic changes that occur in chronic emphysema include:

1. The bronchiolar obstruction *increases airway resistance* and results in greatly increased work of breathing
2. The marked loss of alveolar walls greatly *decreases the diffusing capacity* of the lung, which reduces the ability of the lungs to oxygenate the blood and remove carbon dioxide from the blood

3. The obstructive process is frequently much worse in some parts of the lungs than in other parts, so that some portions of the lungs are well ventilated, while other portions are poorly ventilated

This often causes *extremely abnormal ventilation-perfusion ratios*, with a very low V_a/Q in some parts (*physiologic shunt*), resulting in poor aeration of the blood, and very high V_a/Q in other parts (*physiologic dead space*), resulting in wasted ventilation, both effects occurring in the same lungs

4. Loss of large portions of the alveolar walls also decreases the number of pulmonary capillaries through which blood can pass

As a result, the pulmonary vascular resistance often increases markedly, causing *pulmonary hypertension*

This in turn overloads the right side of the heart and frequently causes right-sided heart failure

Chronic emphysema usually progresses slowly over many years

The person develops both hypoxia and hypercapnia because of hypoventilation of many alveoli plus loss of alveolar walls

The net result of all these effects is severe, prolonged, devastating *air hunger* that can last for years until the hypoxia and hypercapnia cause death—a high penalty to pay for smoking

Pneumonia

pneumonia includes any inflammatory condition of the lung in which some or all of the alveoli are filled with fluid and blood cells

A common type of pneumonia is *bacterial pneumonia*

- the disease begins with infection in the alveoli
- the pulmonary membrane becomes inflamed and highly porous so that fluid and even red and white blood cells leak out of the blood into the alveoli

- the infected alveoli become progressively filled with fluid and cells, and the infection spreads by extension of bacteria or virus from alveolus to alveolus
- eventually, large areas of the lungs, sometimes whole lobes or even a whole lung, become “consolidated,” which means that they are filled with fluid and cellular debris

Two major pulmonary abnormalities that occur in pneumonia:

(1) reduction in the total available surface area of the respiratory membrane and

(2) decreased ventilation perfusion ratio

these effects cause *hypoxemia* (low blood oxygen) and *hypercapnia* (high blood carbon dioxide)

Asthma

Asthma is characterized by spastic contraction of the smooth muscle in the bronchioles, which partially obstructs the bronchioles and causes extremely difficult breathing

It occurs in 3 to 5 % of all people at some time in life

The usual cause of asthma is contractile hypersensitivity of the bronchioles in response to foreign substances in the air

The allergic reaction that occurs in the allergic type of asthma is believed to occur in the following way:

Allergic person has a tendency to form abnormally large amounts of IgE antibodies, and these antibodies cause allergic reactions when they react with the specific antigens that have caused them to develop in the first place

These *antibodies* are mainly attached to mast cells that are present in the lung interstitium in close association with the bronchioles and small bronchi

When the asthmatic person breathes in pollen to which he or she is sensitive (that is, to which the person has developed IgE antibodies), the pollen reacts with the mast cell–attached antibodies and causes the mast cells to release several different substances

Among them are:

- a) *histamine*
- b) *slow-reacting substance of anaphylaxis*
(which is a mixture of leukotrienes)
- c) *Eosinophilic chemotactic factor* and
- d) *bradykinin*

The combined effects of all these factors, especially the slow-reacting substance of anaphylaxis, are to produce:

- (1) localized edema in the walls of the small bronchioles, as well as secretion of thick mucus into the bronchiolar lumens, and
- (2) spasm of the bronchiolar smooth muscle

Therefore, the airway resistance increases greatly

The bronchiolar diameter becomes more reduced during expiration than during inspiration in asthma this is caused by bronchiolar collapse during expiratory effort that compresses the outsides of the bronchioles

Because the bronchioles of the asthmatic lungs are already partially occluded, further occlusion resulting from the external pressure creates especially severe obstruction during expiration

That is, the asthmatic person often can inspire quite adequately but has great difficulty expiring

Clinical measurements show

- (1) greatly reduced maximum expiratory flow rate and
- (2) reduced timed expiratory volume

Also, all of this together results in dyspnea, or “air hunger”

The *functional residual capacity* and *residual volume* of the lung become especially increased during the acute asthmatic attack because of the difficulty in expiring air from the lungs

Atelectasis

Atelectasis means collapse of the alveoli

It can occur in localized areas of a lung or in an entire lung

Its most common causes are:

- 1) total obstruction of the airway
- 2) lack of surfactant in the fluids lining the alveoli